

A Comprehensive Review of Theories, Models and Approaches Governing Climate Change Discourse and Global Climate Finance

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Abstract

This paper presents an interdisciplinary review of major theories, models, and approaches governing climate change discourse and the evolving global climate finance architecture. Drawing from diverse philosophical traditions and interdisciplinary frameworks, the study highlights that climate change is not solely an environmental issue but a complex systemic crisis with profound political, economic, ethical, and epistemological dimensions. The review unveils that no single theory can sufficiently address the multifaceted nature of climate change or offer universally applicable solutions. Instead, an integrated approach bridging macro-structural paradigms with normative and identity-centered frameworks is essential. It further underscores that existing climate finance is neither neutral nor purely technical; it is shaped by power dynamics, historical inequalities, and contested definitions that influence whose priorities are funded and how. It argues that a just and transformative climate action requires reframing these frameworks to better reflect systemic interdependencies, structural inequalities, and diverse contexts.

Despite the proliferation of theoretical and institutional frameworks, several gaps persist: Western epistemic dominance, theory - practice disconnects, definitional ambiguities in climate finance, and fragmentation in governance. To address these, the paper recommends a shift toward inclusive, interdisciplinary, and grounded models. It advocates theorizing from the Global South, integrating ethical and operational frameworks, developing robust metrics for climate finance accountability, scaling just transitions, and localizing global finance mechanisms. In conclusion, the paper asserts that theories, models, and approaches are not merely academic abstractions; they govern the political economy of climate action. As climate risks intensify, bridging theory and practice is imperative for fostering equitable, coherent, and transformative climate governance.

Keywords: Climate finance, Climate change, Climate governance, Climate action, Transformative enablers

Introduction

Expanding climate change impacts and shrinking climate finance are among the most pressing challenges of our time. As climate risks intensify across ecological, economic, and political domains, there is a critical need to revisit the theoretical foundations shaping how these issues are understood, financed, and governed.

This synthesis presents a comprehensive analysis of the theoretical, conceptual, and historical foundations that shape global climate change discourse and the architecture of climate finance. It argues for a reimagining of both domains through integrated, justice-oriented, and forward-looking lenses.

Various disciplines, including humanities and social science, philosophy, implementation science, climate change, and development, are structured around specific theories, models, and approaches. Numerous scholars within these disciplines have emphasized the crucial role of theories, models, and approaches in shaping academic discourse and understanding and directing research, policy, and practice. Prominent figures from each of these disciplines, such as Thomas Kuhn in science (Kuhn, 1962), Immanuel Kant in philosophy (Kant, 1781) and Max Weber in social science (Weber, 1978), Amartya Sen in development (Sen, 1999), and the Intergovernmental Panel on Climate Change (IPCC) in climate science (IPCC 2021), have highlighted how their disciplines are structured around specific theories, models, and approaches that guide inquiry, understanding, and action across disciplines.

Understanding Theories, Models and Approaches

Theories, models, and approaches offer insights into how knowledge systems are organized across different disciplines. For instance, in his influential book *The Structure of Scientific Revolutions* (Kuhn, 1962), Thomas Kuhn proposed that scientific progress occurs through sets of theories and practices known as paradigms, which dictate how research is conducted. Kuhn's notion of 'paradigm shifts' demonstrates how theoretical frameworks govern scientific disciplines, guiding research and data interpretation. He argues that scientific advancement happens when an existing paradigm is supplanted by a new one, based on different theories and models. Similarly, the philosopher Immanuel Kant (1781) argued that human understanding is shaped by a priori theoretical frameworks. In his *Critique of Pure Reason*, Kant illustrates that theoretical frameworks are necessary for understanding and inquiry. He further elaborates that theoretical models structure knowledge allowing us to make sense of the world by setting the boundaries for what can be known and how knowledge is generated. The renowned social scientist, Max Weber, in *Economy and Society* (1978), highlighted that social

science is built on theoretical constructs that help explain human behaviour and societal structures. He argued that social inquiry is governed by theoretical tools that shape the questions asked and the methods used to analyse social systems. The eminent sociologist Emile Durkheim argued that social sciences depend on distinct theoretical frameworks that differentiate them from the natural sciences. In his work *The Rules of Sociological Method* (1895), Durkheim contended that societies must be understood through specific models and theoretical principles that explain social facts and collective behaviours.

Amartya Sen, in his influential book *Development as Freedom* (1999), illustrates that development is governed by theoretical frameworks that determine how progress is measured and understood. Sen's capability approach replaced earlier models that focused solely on economic growth, emphasizing the importance of theoretical models that encompass human well-being, freedoms, and capabilities. Dependency theory, articulated by scholars such as Prebisch (1950) and Frank (1967), governed much of the discourse on development in the mid-20th century. This theory provided a counter-model to modernization theory, arguing that development was structured by global systems of economic dependency and inequality. Theories like this have led to a re-evaluation of development models, emphasizing practitioners to consider the role of global economic systems in shaping local conditions. This approach informs contemporary debates on development aid, trade policies, and global justice, emphasizing the need for equitable economic relationships. Not the least, systems theories such as Earth System Science (Kump, Kasting & Crane, 2010) which views the Earth as an integrated system with feedback loops between the atmosphere, biosphere, lithosphere, and hydrosphere, govern much of climate science research (Rockstrom et al. 2009; Steffen, Crutzen & McNeill, 2007). Fundamentally, the systems theory approach helps scientists understand the interconnections between human actions and climate change (Falkowski & Godfrey, 2021).

Fundamentally, while the concepts of theories, models, and approaches are related and interconnected, they differ in scope and purpose, and hence possess distinct meanings, purposes and roles in research and practice (Moutinho & Phillips, 2022; Babbie, 2021). Each contribute to understanding, formulation, and application of research into practice (Nilsen, 2022, Nilsen, 2020). Comprehending the distinctions among theories, models, and approaches is essential in academic discourse, research, and application (Tabak et al., 2018) as each serves a specific purpose in developing knowledge and its application within any discipline (Creswell, 2014; Cleland, 2001). These concepts are fundamental as they help in identifying contextually relevant frameworks for research into practice (Ngai et al., 2014; Fidel, 2012).

Theories

In essence, theories adhere to a specific research philosophy, or a combination of philosophies based on their objective, focus on explanation, evidence, and conceptual relationships (Denzin & Lincoln 2018; Bryman, 2016). For instance, theories based on positivist research philosophy emphasize the utilization of empirical data and scientific methods to study social phenomena objectively. Positivist theories focus on observable facts and seek to establish generalizable laws or principles that can explain social phenomena (Cohen & Morrison, 2018). Conversely, theories with a realist research philosophy posit that social phenomena possess objective reality, and that the world exists independently of human observers. Realist theories aim to describe the world as it is, rather than as people perceive it (Bhaskar, 2018). In contrast, theories based on interpretivism emphasize the significance of understanding the meaning that individuals attribute to their actions and behaviour. Interpretivist theories focus on subjective meanings and the importance of context in shaping social behaviour (Crotty, 2020), while the pragmatism philosophy of research emphasizes practicality and effectiveness in social research. Pragmatist theories focus on identifying solutions to practical problems and seek to balance theoretical and empirical approaches (Johnson & Onwuegbuzie 2020).

Each research philosophy informs how theory is developed, applied, and understood, reflecting diverse approaches and shaping theoretical frameworks and understanding diverse methodologies in the context of knowledge formulation, creation, and use (Hesse-Biber & Leavy, 2020; Creswell & Poth 2018). Therefore, it is crucial to carefully consider the underlying philosophy of a theory and its implications for research design and interpretation of results (Bourke, 2021; Guba & Lincoln, 2019).

The definition of theory varies across disciplines, as different scholars and theorists emphasize distinct aspects of its function and purpose. Walker and Avant (2011) define a theory as a set of concepts and propositions that provide an organized way of thinking about a phenomenon. This perspective underscores the role of theories in structuring complex ideas, thereby facilitating a deeper understanding of phenomena by linking concepts through established relationships and propositions. Their view of theory as an organized way of thinking about a phenomenon can be seen as consistent with interpretivist philosophy of research, particularly in the context of social sciences as they focus on understanding multiple perspectives and meanings within human interactions, rather than seeking objective predictions. In contrast, Colquitt, J.A., and Zapata-Phelan, C.P. (2007) characterize theories as structured explanations that clarify relationships between constructs, helping to elucidate why certain phenomena occur. This definition emphasizes the explanatory power of theories in relation to observable

events. Their focus on the relationships between constructs and their ability to explain why phenomena occur suggests alignment with realism as realism is concerned with explaining the underlying structures that govern observable events. By focusing on the causal mechanisms that underlie constructs, their definition of theory supports the realist idea that theories help reveal the deeper, often unseen, structures that influence reality. Bacharach (1989) further expands on this notion, positing that a theory is a comprehensive explanation of a fundamental aspect of the natural or social world that is supported by empirical evidence. He stresses that effective theories should elucidate relationships among concepts while also accounting for the constraints and assumptions inherent in their application. He emphasizes that theories are comprehensive explanations supported by a body of evidence. His focus on empirical support and the relationship among concepts suggests a positivist approach, where the goal is to objectively explain aspects of the natural or social world through testable propositions. Similarly, following a positivist research philosophy, Bourke (2021) defines theories as systematic frameworks used to explain observations and relationships among various aspects of life. He suggests that theories are essential for explaining why things happen in a specific way, a particular phenomenon such as delinquency, social stratification, or political behaviours. He further emphasizes that theories not only help identify patterns but also provide logical explanations for these patterns, helping researchers make sense of observed social regularities. Furthermore, theories guide research by directing attention to likely patterns of interest. For example, a theory may hypothesize that social class and delinquency are inversely related (as one rises, the other falls), and this guides how data is collected and analysed. Anol Bhattacharjee (2019, 2012) offers a contemporary perspective, defining a theory as a set of systematically interrelated concepts, definitions, and propositions intended to explain or predict phenomena. In this context, Bhattacharjee highlights that theories serve as frameworks for understanding and anticipating how specific aspects of reality function, achieved by linking various concepts and delineating their relationships. Bhattacharjee's emphasis on systematic relationships and prediction reflects positivism's emphasis on observable facts and cause-and-effect relationships. Therefore, this definition of theory also resonates with positivist philosophy.

The concept of theory exhibits variation across disciplines, with scholars emphasizing distinct aspects of its function. Social theories constitute sets of concepts and propositions that structure complex ideas, aligning with interpretivist philosophy by focusing on the comprehension of diverse perspectives in social interactions. They also function as structured explanations elucidating relationships between constructs, reflecting a realist approach by emphasizing

causal mechanisms underlying observable phenomena. Scientific theories offer comprehensive explanations supported by empirical evidence, consistent with positivism's emphasis on objective, testable propositions. They provide systematic frameworks for explicating social patterns and guiding research by identifying key relationships, such as between social class and delinquency, reinforcing the positivist focus on cause-and-effect relationships and systematic predictions.

Climate science and climate finance theories provide essential frameworks for comprehending and tackling climate change (Zehr, 2014). These theories function as organized systems of ideas and principles that elucidate climate-related phenomena, their consequences, and the financial strategies associated with climate change mitigation and adaptation (Internet Encyclopaedia of Philosophy, 2024). Climate science theory combines insights from various fields, including atmospheric physics, oceanography, biogeochemistry, and ecology, to explain climatic events such as global warming, ocean acidification, and extreme weather. The Anthropogenic Climate Change Theory is backed by evidence from climate models, paleoclimate data, and observational records suggesting that human activities, especially greenhouse gas emissions, are the main cause of recent global warming (IPCC, 2021). In parallel, climate finance theories explore how financial systems can be utilized to address climate change (Grantham Research Institute on Climate Change and the Environment, 2023). The theory of Carbon Pricing, for instance, explains how putting a price on carbon emissions through taxes or cap-and-trade systems can encourage reductions in greenhouse gas emissions by positioning economic incentives with environmental outcomes (Stern, 2007). Another example is the Green Finance Theory, which explores how financial instruments, such as green bonds and climate funds mobilize finances for climate resilience development (Campiglio, 2016). In this manner, climate science and climate finance theories provide a coherent structure for understanding complex interactions within the climate system and the climate economics that respond to or influence climate change discourse.

As in other disciplines, theory building in climate sciences and climate finance is an iterative process that involves developing, refining, and validating theories through systematic research, observation, and analysis. This process integrates empirical data, modelling, and conceptual frameworks to enhance understanding and generate actionable insights. For example, the development of the Planetary Boundaries Theory (Rockstrom et al., 2009) was developed by integrating data on biogeochemical cycles, biodiversity loss, and climate change to define safe operating limits for human activity on Earth. This theory has been refined over time through ongoing research and empirical validation. Similarly, the Climate Risk Theory in finance has

been shaped by empirical studies on the financial impacts of climate change and the development of frameworks for climate risk disclosure, which evolved through research on how physical risks, such as, climate induced disasters, extreme weather and transition risks, such as, policy changes affect financial markets and asset valuations (Task Force on Climate-related Financial Disclosures, 2017).

Models

A model, according to Fried (2020), is a more concrete instantiation of a theory. Models are narrower in scope and serve to represent specific aspects of a theory, providing detailed descriptions of how certain processes or phenomena operate. Fried uses the metaphor of maps, much like a map simplifies a city to aid navigation. Models break down complex processes into their component parts, relationships, and dynamics, and serve as intermediaries between broad theories and the real world. Fried argues that models may be verbal, mathematical, or physical and are useful in testing and illustrating the mechanisms proposed by a theory. Babbie (2020) defines models from a scientific inquiry perspective. He suggests that models are simplified versions of reality that help describe phenomena through a structured framework. According to Babbie, models are representations, often mathematical or conceptual, used to organize and codify the observations useful for testing predictions and determining the accuracy of theories. Models differ from theories in a way that models focus more on the systematic and observable relationships than providing contextualized explanations. For example, a model can represent the relationship between hours spent studying and exam grades. However, it is the theory that seeks to explain why studying more lead to better or a certain delineated grade. Interdisciplinary models in social science, climate finance, and development are evolving. Models evolving in these disciplines serve as important tools for better understanding complex interactions between development, environment and societies (Chirambo, 2020; Monasterolo et al., 2019).

Approaches

Babbie (2020) defines a research approach as a comprehensive framework or methodology for examining specific socioeconomic phenomena. The selected approach may vary based on the epistemological or theoretical foundations guiding the study (Bryman, 2016). Creswell, J. W. (2014) demonstrates how qualitative, quantitative, and mixed-method research approaches are influenced by their underlying epistemology and theoretical assumptions. Furthermore, the grounded theory approach emphasizes developing theory inductively from primary or

secondary data, rather than initiating with a hypothesis derived from existing theory (Charmaz, 2006; Strauss & Corbin, 1990; Glaser & Strauss, 1967). Fried (2020) asserts that an approach differs from a theory or model, as it typically refers to the broader methodology or perspective utilized in research. It encompasses the methods and strategies employed to investigate and apply theories and models. For instance, psychological or scientific approaches involve testing theory-derived hypotheses through experiments, data modelling, or observations. Consequently, an approach includes the practical steps taken to operationalize theories and models within a research framework (Nilson, 2022).

Theories, models, and approaches are interconnected in shaping knowledge, structuring research, and guiding practical application. Theories provide broad frameworks explaining phenomena through conceptual relationships, mostly grounded in different philosophical perspectives like positivism, realism, or interpretivism. Models translate theories into tangible, structured representations, illustrating how theoretical mechanisms operate in a simplified, testable format. Approaches define methodological strategies for applying theories and models, guiding data collection, analysis, and interpretation. In climate science and finance, for example, the Anthropogenic Climate Change Theory explains human-induced climate change, models like climate simulations quantify predictions, and research approaches determine how empirical evidence validates these models. Together, theories generate foundational knowledge, models refine and represent theoretical concepts, and approaches facilitate their systematic investigation and application.

Climate Change Defined

The Intergovernmental Panel on Climate Change (IPCC) defines climate change as a quantifiable alteration in the climate's state or variability that persists for an extended period, typically spanning decades or longer (IPCC, 2014). This phenomenon encompasses climatic shifts over time, whether attributed to natural factors or anthropogenic activities (IPCC, 2018). According to the United Nations (UN), climate change refers to long-term shifts in temperatures and weather patterns, primarily driven by human activities such as burning fossil fuels, which increase greenhouse gas emissions. (United Nations, n.d.). According to NASA, climate change refers to long-term changes in Earth's temperature and weather patterns, primarily caused by human activities, especially the burning of fossil fuels. These activities release greenhouse gases, such as carbon dioxide, which trap heat in the atmosphere and lead to global warming (NASA, n.d.). According to the United Nations Development Programme (UNDP), climate change refers to long-term alterations in temperature, precipitation, and other

atmospheric conditions caused by natural processes and human activities, particularly greenhouse gas emissions from burning fossil fuels and deforestation (UNDP, 2011). These definitions also encapsulate the fundamental idea of climate change. Climate change researchers and scientists have investigated its geopolitical implications, socioeconomic consequences affecting human populations, and impacts on Earth's ecosystems and natural habitats (Nordhaus, 2017). They have projected potential future climate scenarios based on diverse assumptions regarding greenhouse gas emissions and possible risks associated with global warming outcomes (Steffen et al., 2018).

Academics have examined different target levels of atmospheric carbon dioxide to mitigate climate change impacts. They have also discussed the necessity of reducing greenhouse gas concentrations in the atmosphere to avert consequences (Rockstrom et al., 2009; Hansen et al., 2008).

The historical perspective on climate change has evolved through scientific knowledge, cross-disciplinary research, technological advancements, and international collaboration (Weart, 2008). This perspective involves understanding Earth's climate variability and recognizing anthropogenic influence on the climate system. It emphasizes interdisciplinary studies, systematic data collection and analysis, and the integration of scientific research, policy formulation, and practical interventions to address climate change consequences (Oreskes & Conway, 2010).

Contemporary research has advanced climate change studies through interdisciplinary collaboration, technological innovations, and global partnerships. These efforts have enhanced our understanding of Earth's climatic fluctuations and the anthropogenic impact on the climate system (Knight, 2025). A recent study by Zhang & Wang (2023) investigated the influence of scientific collaboration on interdisciplinarity in climate change research, highlighting the importance of multidisciplinary approaches in addressing complex climate challenges. Their research shows that collaborative efforts, both among authors and at the national and international or regional level, increase disciplinary diversity, promoting a comprehensive understanding of climate change complexities.

The emergence of climate change theory, models and approaches

Climate change theory is not a one-time discovery. Rather, it represents a culmination of centuries of calamities, observations, hypotheses, and refinements. The evolution of knowledge and understanding regarding climate change theory can be better understood through a

chronological framework utilizing the following pentachromic segmentation of major developments (Aghel, 2021).

The period from the 18th century to the current times saw significant advancements in climate science, though contributions from various scientists and philosophers. The era before mid-18th century is less prominently documented. This may be due to sociopolitical changes, colonial disruptions, and shifts in scientific centers to Europe and North America. However, there are notable contributions to the fields of astronomy, the science of meteorology, hydrology and environment. Many scholars validate this intersection of sociopolitical history, colonial disruptions, and scientific contributions, for instance, from Muslim majority regions in shaping climate related knowledge (Mikhail, 2011; Hodge, 2007; Ihsanoglu, 2004; Grove, 1997; Salam, 1987; and El-Baz, 1984). These advancements collectively played a crucial role in shaping climate change theory, with various key scientific contributions laying foundation for our current understanding of the anthropogenic effects of climate change (Heymann, 2010).

Table 1: Chronological history of climate change

Chronology: 19 th century (Early observations)
Major highlights ▪ Joseph Fourier's Greenhouse Hypothesis: In the 1820s, French mathematician and physicist Joseph Fourier coined the phenomena of the greenhouse effect and suggested that Earth's atmosphere traps heat, making the planet warmer than it would be without it (Fourier, 1824; Fourier, 1827). ▪ Eunice Newton Foote's Experimental Observations: In 1856, American scientist Eunice Newton Foote demonstrated that carbon dioxide and water vapor absorb heat, suggesting that increased levels of these gases could elevate Earth's temperature (Jackson, 2020; Raymond, 2020). ▪ John Tyndall's Infrared Absorption Studies: In the 1860s, John Tyndall identified gases like water vapor and carbon dioxide (CO₂) as key heat absorbers (Tyndall, 1961). ▪ James Croll's Astronomical Climate Theory: In the 1860s, Scottish scientist James Croll developed a theory linking Earth's orbital variations to climate changes. He suggested that shifts in the planet's orbit trigger ice ages by affecting the distribution of solar radiation on Earth. Croll's work introduced Astronomical Climate Theory, an idea that external astronomical factors could influence Earth's climate over geological timescales (Croll, 1875).
Chronology: Between late 19 th century and early 20 th century. (Emergence of Anthropogenic climate change)

Major highlights: Svante Arrhenius's Greenhouse Gas Calculations: In 1896, Swedish scientist Svante Arrhenius theorized that CO₂ emissions from burning fossil fuels could lead to global warming. His calculations were remarkably close to modern estimates. He conducted the first quantitative analysis of the impact of atmospheric carbon dioxide on Earth's temperature. He calculated that doubling the concentration of CO₂ could lead to a significant increase in global temperatures (Arrhenius, 1896).

Chronology: 20th century (Growing evidence)

Major highlights

- Thomas Chrowder Chamberlin's Climate Hypotheses: In 1899, American geologist Thomas Chrowder Chamberlin expanded on Arrhenius's work by exploring how variations in atmospheric CO₂ concentrations could drive climate changes, including glacial and interglacial periods (Brogdon, 1995).
- Greenhouse Effect Terminology: In 1901, Swedish meteorologist Nils Gustaf Ekholm introduced the term 'greenhouse effect' to describe the warming influence of atmospheric gases like CO₂ and water vapor, highlighting the potential for human-induced climate alterations (Ekholm, 1901).
- Guy Stewart Callendar's Empirical Evidence: During the 1930s, Guy Callendar provided empirical data showing rising CO₂ levels and warming temperatures. He linked this trend to rising CO₂ levels from industrial activities, providing empirical support for the theory of anthropogenic global warming (Callendar, 1938).
- The Keeling Curve: In the 1950s, Charles Keeling began measuring atmospheric CO₂ at Mauna Loa, revealing a steady rise in concentrations (Keeling, 1960).
- In the 1967, researchers Syukuro Manabe and Richard Wetherald developed world's first accurate computer model of Earth's climate predicting that doubling CO₂ concentrations could rise global temperature by approximately 2°C (Manabe & Wetherald 1967).

Chronology: Between late 20th and early 21st century.
(Climate models and global consensus)

Major highlights

- By the 1980s, climate models became more sophisticated, showing clear links between CO₂ and temperature rise (Hansen et al., 1988).
- In 1988, the Intergovernmental Panel on Climate Change (IPCC) was established, providing systematic assessments of climate change science.
- IPCC Assessment Reports (First Report in 1990, latest Sixth Assessment Synthesis Report released on 20th March 2023).

Chronology: 21 st century (Present day understanding)
Major highlights The consensus today is that human activities, especially fossil fuel combustion and deforestation, are the primary drivers of climate change. Extreme weather events, sea-level rise, and biodiversity loss are now linked to anthropogenic climate change (NASA, 2023; WMO, 2023; Cook et al., 2016).

Overview of Major Theories Governing Climate Change Discourse

Climate change discourse is multidisciplinary, multifaceted and multisectoral, rendering it one of the most intricate global challenges (IPCC, 2021; Diaz et al., 2020). It encompasses a wide range of interconnected issues such as extreme weather events, biodiversity loss, food insecurity, water scarcity, public health risks, economic disruptions, and social inequalities (FAO, 2021; Watts et al., 2021; Ostrom, 2020). These challenges manifest differently across regions, ecosystems, and communities, necessitating diverse and context-specific responses (Adger et al., 2020; Hallegatte et al., 2020; Shue, 2021).

The multidisciplinary nature of climate change arises necessity of integrating knowledge from multiple fields (Rockstrom et al., 2021). For instance, environmental science provides critical insights into climate systems, carbon cycles, and ecosystem dynamics. Economics and finance contribute to the design of sustainable financing mechanisms and climate adaptation investment strategies (Atteridge & Canales, 2021; Stern, 2020).

The multifaceted and multisectoral nature of climate change also necessitates coordinated action across various sectors (IRENA, 2021; UNEP, 2021). For instance, the energy sector supports transitioning to renewable sources and reducing emissions in agriculture and food sectors. Finance and governance help establishing climate finance mechanisms, securing adaptation funding, and developing effective policy frameworks (Biermann et al., 2020).

As such, climate change discourse is intricately linked to the domains of politics, economics, security, environment and governance (Dryzek & Pickering, 2020; Barnett & Adger, 2020). Consequently, the narrative surrounding climate change is informed by a diverse array of theoretical frameworks derived from international relations, development studies, environmental governance, sociology, political science, and economics (Biermann et al., 2020). Each theoretical perspective provides distinct insights into the underlying drivers, responses, and rationales for both political action and inaction regarding climate change, while simultaneously illuminating inherent paradoxes and limitations within these frameworks.

The following section maps a broad spectrum of theories, from classical international relations frameworks such as Realism, Liberal Institutionalism, Constructivism to critical perspectives such as Marxist Political Economy, Postcolonial Theory, Feminist and Ecofeminist Thought, and newer paradigms such as Securitization, Systems Theory, Resilience Thinking, and Polycentric Governance, emphasizing their key concepts, inherent paradoxes, and applications within climate regime.

Realism and Climate Politics

Realism, a fundamental theory in international relations, was formalized by Hans Morgenthau in 1948 and expanded by Kenneth Waltz in 1979. This theory asserts that states operate within an anarchic international system, prioritizing national interests and power (Waltz, 1979). Proponents argue that climate change governance is constrained by state-centric considerations, with powerful nations primarily acting to protect their economic and strategic interests rather than pursue global environmental sustainability (Morgenthau, 1948). The withdrawal of the United States from the Paris Agreement during the Trump administration (2017–2021) and its reaffirmation under his second term (2024 – 2028) exemplify realism, as the decision was driven by national interest considerations, prioritizing economic growth over collective climate action (Mearsheimer, 2021). While realism explains the tendency for climate negotiations to result in minimal agreements, it inadequately addresses increasing international cooperation on climate issues (Eckersley, 2004). The paradox in realism regarding climate politics is that, despite climate change being an existential threat to all states, competitive international dynamics hinder collective action. Realism helps explain why states view climate change as a 'threat multiplier', exacerbating geopolitical tensions, such as resource conflicts and migration crises (Methmann & Rothe, 2012). However, it provides limited insight into cooperative governance mechanisms like the Paris Agreement.

Liberal Institutionalism and Regime Theory

Liberal institutionalism, articulated by Robert Keohane in 1984, theorises that international cooperation can be achieved through institutions that mitigate the anarchic nature of the international system and foster collaboration (Keohane, 1984). Climate governance illustrates this framework, with mechanisms like the United Nations Framework Convention on Climate Change (UNFCCC) and the Paris Agreement establishing guidelines and incentives for collective action (Young, 1994). The European Union's leadership in the enforcement of the Paris Agreement, characterized by legally binding commitments and financial mechanisms,

exemplifies the role of institutions in facilitating climate governance, even in the face of challenges posed by state sovereignty (Oberthur & Groen, 2021).

Regime theory further explains the development of climate governance structures in response to shared global challenges. However, a paradox exists in the voluntary nature of climate agreements. While institutions provide coordination mechanisms, they lack enforcement capabilities, leading to weak commitments and noncompliance (Ostrom, 1990). Furthermore, reliance on market-oriented solutions, such as carbon trading, reveals neoliberal biases that prioritize economic efficiency over social equity.

Constructivism – The Role of Norms

Constructivism, as articulated by Alexander Wendt in 1999, posits that social constructs, such as ideas, norms, and identities, significantly influence international relations, superseding material interests (Wendt, 1999). The discourse surrounding climate change has been notably shaped by evolving norms, including the recognition of climate justice and sustainability (Roberts & Parks, 2020; Finnemore & Sikkink, 1998). The constructivist framework elucidates the rationale behind states adopting ambitious climate policies, despite limited material incentives, as evidenced by the ascendance of the ‘net-zero’ narrative and the integration of climate action within the broader context of global governance (Falkner, 2021; Wendt, 2020). The increasing corporate commitments to achieving net-zero emissions, exemplified by Microsoft's 2020 pledge to attain carbon negativity by 2030, illustrate the influence of evolving norms on policy and business practices (Microsoft, 2020). The paradox in constructivism is that while discourse and framing may inform policy decisions, they do not inherently ensure effective implementation. Competing narratives, such as, climate scepticism and economic growth imperatives, have the potential to undermine global consensus and impede timely action (Lewandowsky et al., 2020; Dunlap & McCright, 2020). Furthermore, constructivism encounters challenges in addressing the power dynamics that facilitate the predominance of certain narratives while marginalizing alternative viewpoints, including indigenous perspectives on climate justice (Barnett & Duvall, 2020; Eckersley, 2020).

Marxist and Critical Political Economy Theories

Marxist and critical political economy theories, rooted in Karl Marx's works and expanded by scholars like Foster (2000), assert that capitalism is the fundamental cause of climate change, as the pursuit of profit drives environmental exploitation (Foster, 2000). These frameworks emphasize the role of multinational corporations and neoliberal policies in worsening climate

crises by prioritizing economic growth over ecological sustainability (Harvey, 2020; Newell & Paterson, 2020). The critique of carbon markets as a mechanism allowing corporations to persist in polluting while purchasing offsets underscores the way capitalism influences climate policies, often prioritizing profit over genuine sustainability (Lohmann, 2021). The concept of ‘climate imperialism’ suggests that developed nations externalize environmental costs to the Global South, reinforcing historical inequalities (Bond, 2012). A paradox within Marxist climate analysis is that, while it identifies capitalism as the structural cause of climate change, implementing practical alternatives is challenging within the current global economic order. Initiatives like ‘green capitalism’ and carbon markets often reinforce the economic structures that drive emissions, resulting in contradictions between reformist and transformative approaches (Lohmann, 2021; Newell & Paterson, 2020).

Postcolonial Theory and Climate Justice

Postcolonial theory, which originated from the seminal work of Fanon (1961) and expanded by scholars such as Said (1978) and Chakrabarty (2009), critically examines Western perspectives in climate governance, positing that historical colonial structures significantly influence contemporary climate inequalities (Nixon, 2011). This theory underscores the disproportionate effects of climate change on the Global South, where nations with minimal historical emissions endure the most severe consequences (Chakrabarty, 2009). The discourse at COP27 regarding the Loss and Damage Fund, characterized by developing nations' calls for reparations from industrialized countries, serves as a pertinent illustration of postcolonial arguments advocating for climate justice financing. Proponents of postcolonial scholarship advocate for climate reparations and the necessity of equitable decision-making processes in international negotiations (Grove, 2021). A notable paradox in postcolonial climate discourse is the advocacy for decolonized methodologies alongside the reality that global climate governance is predominantly dominated by Western institutions. While concepts such as ‘climate refugees’ and ‘loss and damage’ serve to interrogate prevailing narratives, their implementation is hindered by persistent power imbalances within global governance structures (Barnett & Duvall, 2020; Nixon, 2011).

Feminist Theory and Ecofeminism

Feminist and ecofeminist theories, emerged in the 1970s with scholars like (Merchant, 2020) and expanded by (Gaard, 2015), examine the gendered dimensions of climate change, highlighting that woman, particularly in the Global South, face greater vulnerabilities due to

systemic inequalities (Gaard, 2015). These frameworks advocate for gender-responsive climate policies and recognize the importance of women's leadership in sustainability movements (MacGregor, 2010). The inclusion of gender-responsive policies in the UNFCCC Gender Action Plan (2019) demonstrates the increasing recognition of feminist perspectives in climate governance (UNFCCC, 2019). A paradox in feminist climate discourse is the tension between essentialist and intersectional approaches (Arora-Jonsson, 2021). While early ecofeminist thought linked women's experiences to nature, contemporary scholars warn against universalizing gendered impacts without considering intersecting factors like race and class (Warren, 2020; Buckingham & Le Masson, 2017).

Securitization Theory and Climate Security

Securitization theory, introduced by the Copenhagen School's securitization theory, and primarily developed by Buzan, Waever, and de Wilde (1998), frames certain issues as security threats, legitimizing policy measures, where political actors justify extraordinary measures. Recently, climate change has been framed within this securitization paradigm, with military and national security agencies incorporating climate risk into strategic planning (Methmann & Rothe, 2012). While this elevates climate change's political priority, it raises concerns about exceptional policies, such as, militarized responses prioritizing state security over human security (Detraz, 2020). This paradox suggests that while securitization may heighten the urgency of addressing climate change, it can divert attention from collaborative mitigation efforts toward border control and resource conflicts leading to risk management approaches rather than sustainable measures. The 2021 North Atlantic Treaty Organization (NATO) report on climate security, which designates climate change as a significant threat to global stability, illustrates how climate issues are being integrated into national, regional security agendas (NATO, 2021). Similarly, monitoring of Pakistan Brickkilns industry under its National Disaster Management Authority (NDMA) is yet another example from South Asia that exemplifies the integration of environmental pollution issue into national security agenda. Critics argue that the process risks marginalizing vulnerable populations and face challenges to address the structural causes of climate change (Oels, 2021; Dalby, 2021).

Sustainable Development and Ecological Modernization Paradigm

The sustainable development paradigm, initially articulated in the Brundtland Commission report aims to achieve a balance among economic growth, social equity, and environmental protection (World Commission on Environment and Development, 1987). Ecological

modernization theorises that technological innovation and policy reforms have the potential to decouple economic growth from environmental degradation (Sachs, 2015). The European Green Deal, launched in 2020 is one such example that seeks to balance economic growth with sustainability, promoting green jobs and investments while reducing emissions (European Commission, 2020). The major paradox regarding the compatibility of continuous economic growth with climate objectives. Critics contend that ‘green growth’ remains dependent on resource extraction and perpetuates existing inequalities (Gupta & Vegelin, 2016). While sustainability frameworks inform global policies, such as the Sustainable Development Goals (SDGs), the challenge resides in their effective implementation within capitalist economies.

Anthropogenic Climate Change Theory

The theory of anthropogenic climate change, established through the contributions of the Intergovernmental Panel on Climate Change (IPCC), is predicated on seminal scientific work that human activities, most notably fossil fuel combustion, deforestation, and industrial emissions, serve as the principal drivers of global warming (IPCC, 2021; Mann et al., 2020). This theory is substantiated by extensive evidence derived from climate models, atmospheric data, and historical temperature records, which collectively illustrate a robust correlation between greenhouse gas emissions and the increase in global temperatures (NASA, 2021; Petit et al., 2021; Mann et al., 2020). Consequently, policies such as carbon pricing, the adoption of renewable energy, and emissions reductions articulated in international treaties are formulated within this theoretical framework.

The record-breaking heatwaves in Europe in 2022 and recent devastating floods in South Asian nations, which scientists linked directly to human-induced climate change, exemplify the real-time impact of anthropogenic global warming. Attribution studies by the World Weather Attribution group confirmed that these extreme weather events were made significantly more likely due to greenhouse gas emissions. A paradox in this discourse is that despite the overwhelming scientific evidence supporting the notion of human-induced climate change, political and economic interests frequently obstruct decisive action. Climate denialism, motivated by entrenched corporate and ideological interests, persists in undermining policy advancement (Oreskes & Conway, 2010).

Natural Climate Variability (non-anthropogenic) Theory

Scholars also contend that while humans are the predominant cause of climate change, natural variability continues to exert an influence on climate patterns (Deser et al., 2020; Hawkins &

Sutton, 2020). Some theories attribute climate change to natural variability, a perspective examined by climatologists such as Lindzen (2001) and Soon and Baliunas (2003), who suggest that such changes may be partially or entirely due to natural causes, including solar cycles, volcanic activity, and ocean currents (Lindzen, 2001). Proponents of non-anthropogenic climate change theories contend that the Earth's climate has experienced natural fluctuations over millions of years, characterized by periods of warming and cooling that are independent of human influence. They reference historical climate events, such as the Medieval Warm Period and the Little Ice Age, as evidence of this natural variability (Soon & Baliunas, 2003). The 2015–2016 El Nino event caused global temperature spikes, extreme droughts, and heavy rainfall, demonstrating how natural climate variability can influence short-term climate patterns. While human-induced climate change amplifies such phenomena, El Nino and La Nina cycles show that non-anthropogenic factors also play a role in climate variability (NOAA, 2016). However, while it is acknowledged that natural factors do influence climate, the rate and magnitude of current warming are unprecedented in recorded history and cannot be wholly accounted for by natural cycles alone (NASA, 2020). The paradox inherent in non-anthropogenic climate theories is that, despite recognizing climate variability, they frequently downplay scientific findings regarding anthropogenic influence. Consequently, these perspectives have been politically employed to justify inaction on climate mitigation policies (Farrell, 2021; Oreskes & Conway, 2020).

Contingency theory

Contingency theory, first proposed by scholars like Fred Fiedler in the 1960s, postulates that there is no one-size-fits-all approach to decision-making; instead, responses depend on specific situational variables (Fiedler, 1967). When applied to climate change, this theory suggests that governance strategies must adapt to the unique political, economic, and environmental contexts of each situation. Consequently, climate policies that work well in one context may be ineffective in another, highlighting the need for flexible, context-specific strategies (Adger et al., 2020). The differentiated responsibilities outlined in the Paris Agreement illustrate this contingency-based approach, allowing nations to develop Nationally Determined Contributions (NDCs) that reflect their individual capacities and circumstances. While contingency theory promotes adaptive governance, its emphasis on situational variability poses challenges for global coordination, as it is difficult to align standardized international climate frameworks with solutions tailored to local conditions (Biermann et al., 2020; Donaldson, 2020).

Risks and Governmentality Theory

Risk and governmentality theory, as articulated by Foucault (1991) in the late 20th century, scrutinizes the modalities through which power is exercised via the management of risks and uncertainties, rather than through overt coercion. Within the framework of climate change, this theory suggests that governance manifests through mechanisms such as emissions trading, climate adaptation planning, and resilience-building, which effectively transfer responsibility to individuals and institutions, rather than relying on state enforcement (Bulkeley & Stripple, 2020; Dean, 1999). The United Kingdom's Net Zero Strategy (2021) underscores the significance of financial incentives, emissions trading, and corporate responsibility, reflects a risk-based governance approach to climate policy (DBEIS, 2021). The fundamental paradox with this theory is that while it enables flexible and decentralized responses, it simultaneously risks depoliticizing climate change by framing it as a technocratic issue rather than a political one (Oels, 2021).

Hegemony Theory

Hegemony theory, as articulated by Laclau and Mouffe (1985), examines the mechanisms through which dominant ideas and discourses construct social and political realities. Within the context of climate discourse, this theoretical framework explains how certain narratives, such as, the prioritization of economic growth over environmental protection, attain hegemonic status, while alternative viewpoints, including degrowth and indigenous knowledge, are systematically marginalized (Hickel, 2021; Whyte, 2020). The dominance of economic growth narratives in climate discourse, despite increasing calls for degrowth, shows how hegemonic ideas shape policy discussions and priorities (Hickel, 2021). The underlying paradox is that entrenched power structures, as hegemonic narratives regarding climate often align with the interests of powerful actors, rather than addressing the needs of those most adversely affected by climate change (Backstrand & Lovbrand, 2021; Newell & Paterson, 2020).

Apocalypse as a Political Logic

The logic of apocalypse in climate politics, as described by Methmann and Rothe (2012), refers to the framing of climate change as an existential crisis necessitating urgent, large-scale intervention. Media and political narratives surrounding the 2019–2020 Australian bushfires framed the disaster as a harbinger of climate collapse, driving public pressure for stronger policies. While apocalyptic narratives can mobilize action and create a sense of urgency, they also risk fostering fatalism, political paralysis, or justification for authoritarian governance

under the guise of emergency response (Hulme, 2020). This paradox demonstrates the tension between motivating action and avoiding public despair or reactionary policies (Methmann & Rothe, 2012).

Polycentric Governance

Polycentric governance, conceptualized by Ostrom (2010), suggests that climate action should not rely solely on centralized, top-down governance, rather involve multiple, semi-independent actors across different levels of society. This theory argues that local, national, and global initiatives can complement each other, creating a more objective resilient and adaptive governance system (Cole, 2020). The global Cities Race to Zero campaign, where municipal governments independently commit to climate action, exemplifies how decentralized governance contributes to climate resilience (C40 Cities, 2021). However, the paradox of polycentric governance is that while it promotes flexibility and innovation, it also risks fragmentation, inconsistency, and duplication of efforts if coordination mechanisms are weak (Jordan et al., 2021).

Systems theory

Systems theory, originating from the contributions of Bertalanffy (1968) and subsequently applied to climate governance, underscores the interconnectedness of environmental, social, and economic systems. This framework explains how feedback loops, tipping points, and nonlinear interactions complicate climate governance, thereby rendering the prediction of policy outcomes uncertain (Levin et al., 2021; Duit et al., 2010). The cascading effects of the 2021 Texas power crisis, where extreme weather disrupted energy, water, and transportation networks, highlight the interconnected nature of climate systems (Electric Reliability Council of Texas [ERCOT], 2021). A paradox inherent in systems theory is that, while it emphasizes the necessity for holistic approaches, its complexity frequently overwhelms policymakers, resulting in inertia or the implementation of fragmented solutions (Lenton et al., 2020).

Resilience theory

Resilience theory, emerged from ecology in the 1970s through the work of C. S. Holling, focuses on how systems absorb disturbances and adapt to change (Holling, 1973). In climate change discourse, resilience thinking is applied to disaster preparedness, adaptation planning, and sustainable development, encouraging societies to build adaptive capacities rather than merely mitigate risks (Cutter et al., 2021; Folke et al., 2020).

The Climate Resilience and Adaptation Finance and Technology Transfer Facility (CRAFT), launched in 2018 that funds climate adaptation and resilience in developing economies characterises one such example. However, the paradox of resilience theory is that it can sometimes reinforce neoliberal governance by shifting responsibility from states to communities and individuals, emphasizing self-sufficiency over structural transformation. The individualized responsibility for climate risks often masks structural inequalities and the need for systemic change (Joseph, 2020; Walker & Salt, 2006).

Tragedy of commons theory

The tragedy of the commons, introduced by Hardin (1968), describes how individuals acting in their own self-interest can deplete shared resources, leading to collective environmental degradation. This theory is widely used to explain the challenges of climate governance, where nations hesitate to reduce emissions due to concerns about free-riding by others (Barrett, 2021; Ostrom et al., 2020). The overexploitation of global fisheries, despite international regulations and treaties, demonstrates the difficulty of managing common resources without strong enforcement mechanisms (FAO, 2020). The paradox is that while the tragedy of the commons suggests strict regulation is necessary, real-world climate policies often rely on voluntary commitments and market-based solutions, which struggle to prevent overuse of the global carbon budget (Frischmann et al., 2021; Ostrom, 1990).

A comprehensive understanding of climate change requires not only theoretical insights into the various discourses shaping the narrative but also a thorough examination of the mechanisms implemented to achieve climate goals. Among these mechanisms, climate finance emerges as a crucial response, translating theoretical frameworks into actionable strategies. The following discussion explores the complexities of climate finance, elucidating how financial instruments and policies are formulated and how they, in turn, influence the prevailing discourses surrounding climate change.

Climate finance – need for a transformative yet a common definition

Climate finance refers to the financial resources flow mobilized public, private or alternative resources to support climate change adaptation, mitigation, and enabling systematic transitions for upscaling adaptation and mitigation solutions and best practices through research, innovation, policy support and capacity building (GCF, 2023; IPCC 2022; OECD 2021; Climate Policy Initiative, 2021; Buchner et al, 2019). It includes public and private flows, national and international sources, and multiple financial mechanisms (UNFCCC, 2022).

Under the principle of ‘common but differentiated’ responsibilities, climate finance through various modalities are the commitments of developed economies, under the United Nations Framework Convention on Climate Change (UNFCCC) and the Paris Agreement, to developing and least developed economies (OECD, 2023; COP26, 2021). According to (Nakhooda, Watson, & Schalatek, 2016) climate finance concerns the financial resources mobilised to support vulnerable countries adapt to and mitigate the impacts of climate change. These definitions provide a nuanced, forward-looking, and holistic articulation of climate finance that addresses gaps in the traditional definitions. Traditional definitions of climate finance narrowly focus on funding for mitigation and adaptation, often treating them as siloed objectives (UNFCCC 2015). For instance, core objective of UNFCCC’s climate finance suggest that ‘climate finance aims at reducing emissions and enhancing sinks of greenhouse gases and aims at reducing vulnerability of, and maintaining and increasing the resilience of, human and ecological systems to negative climate change impacts. The downside of existence of multiple definitions of climate finance, combined with inconsistent accounting practices, undermines efforts to standardize climate finance and hampers effective coordination across financial instruments (Climate Funds Update & Heinrich Boll Stiftung, 2024).

Definition of climate finance proposed in this synthesis paper explicitly includes ‘enabling systemic transitions’ and ‘upscaling solutions’, recognizing that finance must not only address immediate or short-term climate impacts but also transform systems (energy, agriculture, governance) to sustain long-term resilience. This aligns with the IPCC’s call for ‘systemic transitions’ to achieve net-zero economies (IPCC, 2023).

Traditional definitions overlook ‘enabling activities’ like research, innovation, policy support, and capacity building, treating them as ancillary rather than foundational (OECD, 2021). The definition introduced in this paper integrates ‘research, innovation, policy support, and capacity-building’ as core pillars for effective climate finance. Because research can drive essential breakthroughs in climate-smart technologies, such as, green hydrogen, green batteries. Capacity building enables stakeholders better design and implement climate solutions (Ahmad 2019). Both these aspects align with the Global Environmental Facility’s (GEF), and GCF’s emphasis on ‘readiness programmes’ to support vulnerable nations (GEF, 2022; GCF, 2023).

Classic climate finance frameworks emphasize – often over emphasize, on public finance such as grants, multilateral funds, and private investments, such as, green bonds, neglecting alternative resources, such as, essential informal financial networks and blended finance settings, e.g., India’s National Adaptation Fund as considered in this definition (Ghosh, 2020).

Traditional definition treats climate finance as a static pool for isolated projects rather than a catalyst for scaling proven solutions, e.g., solar microgrids, drought/flood resistant crops. Climate finance definition in this paper aligns with GCF mandate to leveraging impacts at scale through replication and up-and-out scaling. Traditional climate finance frameworks tend to be blind spot – rarely link finance to structural socioeconomic shifts, such as, just transitions for fossil fuel replacement, GEDSI-responsive budgeting. The 'enabling systemic transitions' in this definition implicitly addresses equity and justice which is closely aligned with the emerging just transition principle of the Paris agreement (UNFCCC, 2015).

Climate change definition proposed in this paper is dynamic in the sense that it considers how finance achieves goals through research, innovation and scaling. It is transformative in the sense that it embeds finance within broader societal transitions. It is actionable as it guides policy makers to fund enablers, not just transactional and shorter-term projects. In the face of competing development objectives, this definition calls for expanded scope of climate finance from project funding to systemic change. It centres on equity and prioritizes capacity building and bridges gaps in the sense that it connects climate finance to innovation and policy, addressing the 'know-do gap' in climate action (UNEP, 2022). This definition aligns with the emerging advocacy for climate finance to be 'transformational', not transactional (Buchner et al., 2022).

The Evolution of Climate Finance and Environmental Governance Frameworks

In the aftermath of World War II, the International Monetary Fund (IMF) and the World Bank Group and International Bank for Reconstruction and Development (IBRD) were established at the Bretton Woods Conference in July 1944. Initially, their primary goal was to support the reconstruction of war-torn European economies. However, over time, their focus shifted toward financing global development, particularly infrastructure and economic growth in developing and least developed economies. Despite being created at the same conference, the IMF and IBRD had distinct mandates. The IMF was designed to manage global financial stability by providing short-term urgency loans to countries facing economic crises and offering policy guidance. The World Bank group focused on promoting long-term economic growth by providing loans, primarily for infrastructure development. Simultaneously, in 1945, the newly established United Nations (UN) created the Food and Agriculture Organization (FAO) to address food insecurity, malnutrition, and living standards in vulnerable nations. Recognizing the deep link between food insecurity and rural poverty, the International Fund for Agricultural Development (IFAD) was established in response to the 1974 World Food

Conference. IFAD's primary role was to finance agricultural development, particularly in impoverished rural areas, helping smallholder farmers enhance their livelihoods. In 1960, the World Bank further expanded its efforts to assist the poorest nations by creating the International Development Association (IDA). IDA serves as the Bank's concessional lending arm, offering low-interest or interest-free loans and grants to support socioeconomic development in the world's poorest countries. Elsewhere in Europe, the Nordic Development Fund (NDF) was founded in 1989 to provide soft loans, grants, and concessional financing to Least Developed Countries (LDCs), Other Low-Income Countries (OLICs), and Lower Middle-Income Countries (LMICs). Later, it shifted its focus on financing climate adaptation and mitigation efforts.

These major developments laid the foundation for modern global development-finance models, financing frameworks, and approaches to climate finance (Gupta & Vegelin, 2020). While originally designed for socioeconomic development, these institutions indirectly contributed to the emergence of global environmental governance. They helped create greater awareness of the need for sustainable practices setting precedents for multilateral cooperation, integrating environmental goals into development (Clemencon, 2016). Over time, they helped shape international frameworks, conventions, and agreements that now drive climate action worldwide (Atteridge & Canales, 2021). The implementation of large-scale programs by these major international financial and development institutions unwittingly shaped the global urgency narrative for environmental protection, natural resource management, and conservation (Barnes, 2025).

While these programs aimed to drive economic growth and development, they often exacerbated environmental degradation, resource depletion, and socio-economic vulnerabilities (McMichael, 2020). In many cases, they also contributed to a debt-environment trap, where countries prioritized economic survival over environmental sustainability, further compelling the international community to adopt environmental conventions (Buscher & Fletcher, 2020). For instance, the Structural Adjustment Programs (SAPs) in the 1980s and 1990s required developing nations to liberalize economies, privatize industries, and promote large-scale extractive projects. These policies accelerated deforestation, land degradation, and biodiversity loss, triggering environmental crises that intensified global pressure for stronger environmental governance (Escobar, 2020). They led the World Bank Group to establish environmental and social safeguards in the 1990s.

Similarly, large-scale infrastructure projects, such as dams, transportation networks, and large-scale mining projects funded by these institutions, resulted in widespread deforestation, habitat

destruction, and land degradation (Adams, 2021). At the same time, the Green Revolution strategies (1950s–1980s) promoted through FAO, IFAD and other agencies to enhance food security also led to unintended environmental impacts (Patel, 2021). Other seminal scholars such as Tilman et al., (2020); Pingali, (2020), and (Khosla & Jat, (2020) highlighted the trade-offs between agricultural intensification during the Green Revolution and its environmental consequences including soil degradation water pollution, biodiversity loss, and increased carbon emissions, reducing crop biodiversity due to heavy reliance on environmentally unfriendly agricultural practices for increasing food security.

The environmental costs of these unintended developments alarmed scientists, environmental activists, and policymakers, creating a strong urgency narrative that ultimately drove the formation and strengthening of global environmental conventions (Andresen & Hey, 2020; Biermann & Kim, 2020; Methmann & Rothe, 2012). The role of these institutions in financing large-scale infrastructure, resource extraction, and unsustainable agricultural practices helped shape international awareness, culminating in key global environmental agreements (Rockström et al., 2021; Meadowcroft, 2020). [Table 1 in Annex-A](#) presents a chronological overview of major global environmental conventions and agreements demonstrating how global environmental governance has evolved overtime in response to pressing environmental challenges such as pollution, biodiversity loss, climate change, and resource degradation.

At the same time, these global developments catalysed orchestrating and formulation of important regional and sub-regional (transboundary) treaties ([please refer to table 1 in Annex B](#)) concerning environmental governance, borderland management, trade cooperation, and water resource allocation in South Asia and Southeast Asia (Islam & Sarker, 2021; Dressler et al., 2021; Barua & Vij, 2020; Suhardiman et al., 2020; Sinha & Bhattacharjee, 2020).

Simultaneously, these global developments, including the rise of environmental awareness also orchestrated formulation of key agreements and initiatives ([please refer to table 2 in Annex - B](#)) on climate change adaptation and mitigation, and climate finance in South Asia and Southeast Asia (Atteridge & Canales, 2021; Islam & Sarker, 2021; Dosch & Hensengerth, 2020; Lebel et al., 2020; Moorthy et al., 2020).

Evolution of Climate Finance – From Political Pledges to Institutionalized Mechanisms

These developments, coupled with escalating threats of climate change, have contributed to the emergence of the ‘triple planetary crisis’ narrative. This, in turn, has necessitated the creation of an expansive governance architecture characterized by a complex web of institutions commonly referred to as ‘climate regime’, and a diverse array of actors operating across

multiple levels (Barnes, 2025; Oberthur & Stokke, 2021; UNEP, 2021; Gupta & Vegelin, 2020; Hale & Roger, 2020). This architecture has ostensibly evolved to address the multifaceted nature of climate change spanning mitigation, adaptation, finance, and associated policies (Bulkeley & Betsill, 2021; Biermann et al., 2020). From this perspective, the formal history of contemporary climate finance is relatively recent, beginning with the establishment of the Global Environment Facility (GEF) in 1991 as a pilot initiative under the World Bank (World Bank, 2021), which was later restructured as an independent entity in 1994. The United Nations Framework Convention on Climate Change (UNFCCC) was adopted in 1992 and entered into force in 1994, with the GEF serving as its financial mechanism (GEF, 2020; UNFCCC 2020). In 2001, three specialized funds were introduced to enhance climate finance: the Least Developed Countries Fund (LDCF) and the Special Climate Change Fund (SCCF), both operating under the GEF (GEF, 2021), and the Adaptation Fund (AF), which was established under the Kyoto Protocol and became operational in 2007 (Mace, 2020).

Following these developments, the formalization of climate finance as a structured global commitment emerged through a series of international agreements under the United Nations Framework Convention on Climate Change (UNFCCC). The Copenhagen Accord, adopted at the 15th Conference of the Parties (COP15) in 2009, marked a significant political milestone in climate finance. While not legally binding, the Accord represented the first major pledge by developed economies to mobilize USD 30 billion in fast-start climate finance for the period 2010–2012 to support climate action in developing and least developed countries. Additionally, it outlined an aspirational goal of mobilizing USD 100 billion annually by 2020 from a combination of public and private sources to assist climate adaptation and mitigation efforts in vulnerable nations (UNFCCC, 2009). Building on this foundation, the Cancun Agreements, adopted at COP16 in 2010, formally incorporated key elements of the Copenhagen Accord into the UNFCCC framework. These agreements reaffirmed the USD 100 billion per year commitment, established the Green Climate Fund (GCF) as a dedicated financial mechanism to channel climate finance, and introduced the Cancun Adaptation Framework to enhance adaptation planning and resilience-building in developing countries (UNFCCC, 2011). The institutionalization of these mechanisms marked a pivotal step in consolidating climate finance governance within the multilateral climate regime.

At COP17 in 2011, the Durban Platform for Enhanced Action was introduced, setting the stage for the development of a new, legally binding framework applicable to all countries (UNFCCC, 2012). This process culminated in the adoption of the Paris Agreement at COP21 in 2015, which aimed to limit global warming to well below 2°C, with efforts to limit it to 1.5°C above

pre-industrial levels. The Paris Agreement reaffirmed the USD 100 billion per year commitment and introduced the Global Stocktake mechanism, designed to assess collective progress toward achieving the agreement's long-term climate goals (Rajamani et al., 2021; UNFCCC, 2015). These developments highlight the evolution of climate finance from political pledges to institutionalized mechanisms within the UNFCCC framework, progressively strengthening global efforts to mobilize financial resources for climate action in developing and least developed countries (Atteridge & Canales, 2021; Biermann et al., 2020).

The Global Climate Finance Architecture

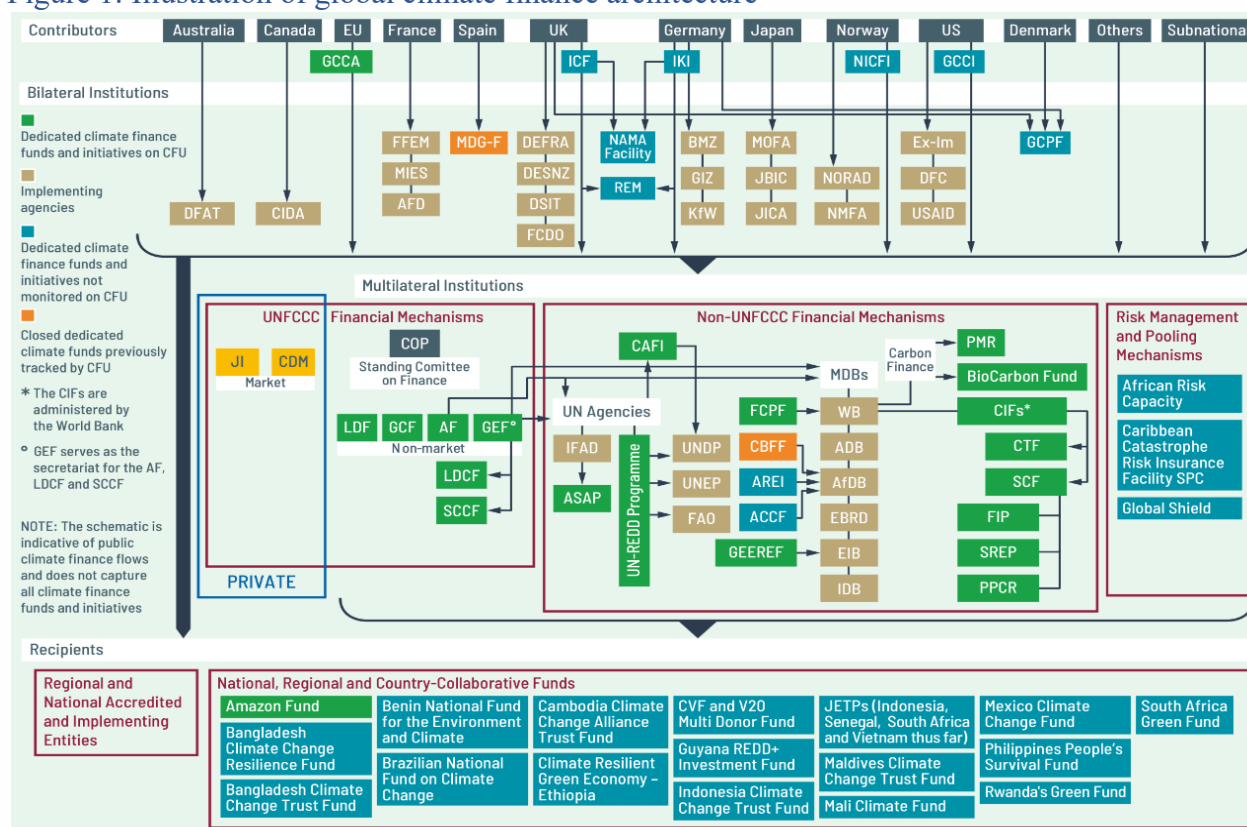
The global climate finance architecture is marked by its inherent complexity and dynamism, reflecting the evolving landscape of international cooperation and financing instruments in response to climate change. Climate finance is distributed through a diverse array of mechanisms, including multilateral frameworks, vertical funds, sovereign wealth funds, regional development banks, and bilateral development assistance, both within and beyond the auspices of the United Nations Framework Convention on Climate Change (Watson et al., 2024). Increasingly, funding is also directed through regional and national climate finance mechanisms established by developing countries, which are actively developing institutional arrangements and collaborative platforms to attract and manage climate-related resources (Atteridge & Canales, 2021; Bhattacharya & Innes, 2020). These mechanisms, specifically dedicated to addressing mitigation and adaptation objectives, remain central to the flow of global climate finance. Concurrently, numerous developed countries have initiated bilateral climate finance initiatives or have integrated climate finance into their development cooperation frameworks. Furthermore, regional and national funds have emerged throughout the Global South, aimed at enhancing country ownership and ensuring greater alignment with national climate priorities (Atteridge & Canales, 2021; Bhattacharya & Innes, 2020). These funds are essential for localizing climate finance, improving accessibility, and tailoring investments to specific socioeconomic and ecological contexts.

Current reform initiatives within multilateral development banks (MDBs) are designed to foster conditions conducive to the advancement of climate finance. These reforms prioritize the mobilization of private capital, the diversification of financing instruments, and the assurance that investments yield substantial benefits for communities without exacerbating debt burdens or compromising future development trajectories (Gabor, 2021; Griffith-Jones & Attridge, 2020). A pivotal aspect of these reforms is the increasing acknowledgment of the necessity to incorporate debt sustainability into the climate finance discourse, thereby ensuring

that climate action does not jeopardize fiscal stability in vulnerable nations (Bhattacharya & Innes, 2020). Additionally, there is a significant shift towards enhancing the technical robustness of Nationally Determined Contributions (NDCs) and National Adaptation Plans (NAPs) to ensure they are not only feasible but also financially viable and appealing to investors (UNFCCC, 2021; Climate Policy Initiative, 2020). This requires a heightened focus on the quality of financial mechanisms, emphasizing concessionality, accessibility, affordability, and transparency, thereby making climate finance more predictable and impactful. These dimensions are crucial for achieving the necessary scale of climate investments required to meet both mitigation and adaptation objectives as outlined in the Paris Agreement.

The Global Climate Finance Architecture 2024 report (see Figure 1) provides a comprehensive, structured yet simplistic schema of the current climate finance landscape. It systematically outlines the institutions, funding mechanisms, and channels through which climate finance is mobilized and disbursed (Watson et al., 2024). The existing climate finance architecture underscores the increasing complexity and fragmentation of the system, as well as persistent challenges related to coordination, transparency, and equitable access, particularly for nations and communities most vulnerable to climate change. For instance, the difficulty in monitoring climate finance flows primarily arises from the absence of an internationally recognized definition of climate finance and the lack of standardized accounting methodologies. These challenges create opportunities for risk management instruments and innovative financing solutions within the evolving climate finance ecosystem and the prevailing mechanisms aim to enhance inclusivity, simplify access procedures, and harmonize tracking mechanisms. While the global climate finance architecture continues to expand in both scope and ambition, its effectiveness depends on the reform of existing institutions, the alignment of investments with national priorities, and the enhancement of the quality and accessibility of financial resources (Buchner et al., 2021). Addressing these structural challenges remains to be essential to convert financial commitments into transformative climate outcomes, especially for least developed and developing nations and those particularly vulnerable to climate change.

Figure 1: Illustration of global climate finance architecture



Implementing Agencies and Institutions	Multilateral Funds and Initiatives	Bilateral Funds and Initiatives
AfDB African Development Bank	ACT Accelerating Coal Transition program (implemented through WB, ADB, AfDB, EBRD and IDB)	GCCI Global Climate Change Initiative (US)
AFD Agence Française de Développement (French development agency)	AF Adaptation Fund (GEF acts as secretariat and WB as trustee)	GCPF Global Climate Partnership Fund (Germany, UK and Denmark)
ADB Asian Development Bank	ACCF Africa Climate Change Fund	ICF International Climate Finance (UK)
BMZ Bundesministerium für Wirtschaftliche Zusammenarbeit und Entwicklung (federal ministry of economic cooperation and development, Germany)	AREI African Renewable Energy Initiative	IKI Internationale Klimaschutzinitiative (international climate initiative, Germany)
CIDA Canadian International Development Agency	ASAP Adaptation for Smallholder Agriculture Programme	MDG-F MDG Achievement Fund (implemented by UNDP)
DESNZ Department for Energy Security and Net Zero (UK)	CAFI Central African Forest Initiative	NAMA Facility Nationally Appropriate Mitigation Action Facility (UK, Germany, Denmark and the EC)
DEFRA Department for Environment, Food and Rural Affairs (UK)	CBFF Congo Basin Forest Fund (hosted by AfDB)	NICFI Norway's International Climate Forest Initiative
DFAT Department of Foreign Affairs and Trade (Australia)	CDM Climate Investment Funds (implemented through WB, ADB, AfDB, EBRD and IDB)	REM REDD+ Early Movers (Germany and UK)
DFC United States International Development Finance Corporation	CIF Clean Development Mechanism (implemented under the Kyoto Protocol)	
DSIT Department for Science, Innovation and Technology (UK)	CTF Clean Technology Fund (implemented through WB, ADB, AfDB, EBRD and IDB)	
EBRD European Bank for Reconstruction and Development	FCPF Forest Carbon Partnership Facility	
EIB European Investment Bank	FIP Forest Investment Program (implemented through WB, ADB, AfDB, EBRD and IDB)	
Ex-Im Export-Import Bank of the United States	GCCA Global Climate Change Alliance	
FAO Food and Agriculture Organization of the United Nations	GCF Green Climate Fund	
FCDO Foreign, Commonwealth and Development Office (UK)	GEF Global Environment Facility	
FFEM Fonds Français pour l'Environnement Mondial (French global environment facility)	GEEREF Global Energy Efficiency and Renewable Energy Fund (hosted by EIB)	
GIZ Deutsche Gesellschaft für Internationale Zusammenarbeit GmbH (German technical cooperation)	JI Joint Implementation (implemented under the Kyoto Protocol)	
IDB Inter-American Development Bank	LDCF Least Developed Countries Fund (hosted by the GEF)	
IFAD International Fund for Agricultural Development	LDF Loss and Damage Fund	
JBIC Japan Bank of International Cooperation	PMR Partnership for Market Readiness	
JICA Japan International Cooperation Agency	PPCR Pilot Program for Climate Resilience (implemented through WB, ADB, AfDB, EBRD and IDB)	
KfW Kreditanstalt für Wiederaufbau (German development bank)	SCCF Special Climate Change Fund (hosted by the GEF)	
MIES Mission Interministérielle de l'Effet de Serre (inter-ministerial taskforce on climate change, France)	SCF Strategic Climate Fund (implemented through WB, ADB, AfDB, EBRD and IDB)	
MOFA Ministry of Foreign Affairs (Japan)	SREP Scaling Up Renewable Energy Program in Low Income Countries (implemented through WB, ADB, AfDB, EBRD and IDB)	
NMFA Norwegian Ministry of Foreign Affairs	UN-REDD Programme United Nations Collaborative Programme on Reducing Emissions from Deforestation and Forest Degradation	
NORAD Norwegian Agency for Development Cooperation		
UNDP United Nations Development Programme		
UNEP United Nations Environment Programme		
USAID United States Agency for International Development		
WB World Bank		

Figure 1 source: Adopted from Global Climate Finance Architecture 2024 (p. 3), by C. Watson, L. Schalatek, & A. Evequoz, 2024, Heinrich Boll Stiftung and ODI. CC BY-NC 4.0.

Traditional Financial Theories and Climate Finance Alternatives

Over the years, climate finance has become a highly debated subject in global policy discourse, and endeavours aimed at addressing climate change (Atteridge & Canales, 2021). With the threats of climate change become increasingly evident, the demand for climate finance has risen substantially underscoring the necessity of sustainable financial mechanisms (UNEP, 2021; Buchner et al., 2020). Despite the growing acknowledgment of its importance, climate finance remains characterized by disparities and fragmentation, shaped by market dynamics, geopolitical socioeconomic structures, and competing crisis narratives (Khan et al., 2020).

Finance scholars have lately engaged with climate finance mechanisms. Historically, they have focused on market efficiency, financial instruments and risk management, largely overlooked inclusive, justice-focused, results-oriented, intersectional and transformative approaches and associated structural aspects requisites of climate finance (Schoenmaker, 2019).

Traditional financial theories, including the Modigliani-Miller Theorem (Heinkel et al., 2020) the Capital Asset Pricing Model (Bolton & Kacperczyk, 2021), Option Pricing Theory (Dixit & Pindyck, 2021), the Efficient Market Hypothesis (Fama & French, 2020), Agency Theory (Jensen & Meckling, 2021), and Pecking Order Theory (Myers & Majluf, 2020) have been applied in the context of climate finance. These theories, along with additional frameworks and models as outlined in [table 1 Annex-C](#), offer essential insights into risk perception, investment behaviour, and capital allocation, thereby aiding policymakers and practitioners in developing alternative climate finance framework (Paseda & Okanya, 2020). However, the sophisticated and distinctive yet advanced and idiosyncratic characteristics of climate change marked by long-term impacts, significant uncertainty, pervasive externalities, and intricate public-private interdependencies render traditional financial models insufficient (Stern, 2020; Weitzman, 2021). As a result, traditional finance theories frequently requisite extension to address the multidimensional challenges inherent in climate finance and to promote more responsive, equitable, and forward-looking financing approaches (Roberts & Weikmans, 2021; Buchner et al., 2020).

Alternative climate finance frameworks have emerged ([please refer to table 1 in Annex C](#)) in response to increasing sophistication, distinctiveness, and idiosyncrasy in climate science and the limitations inherent in traditional finance theories and models, which often fail to adequately address the urgent and context-specific needs of climate finance recipient countries. Operational management of climate finance further nictitated alternative climate finance mechanisms (World Bank, 2021; Buchner et al., 2020). Key factors driving this shift include the protracted and bureaucratic disbursement processes associated with climate regime, the

lack of direct access for local stakeholders, and the misalignment between national priorities and donor-driven funding models (Roberts & Weikmans, 2021; Buchner et al., 2020; Khan et al., 2020). These alternative approaches, developed by think tanks, coalitions, and development agencies, aim to enhance accessibility, flexibility, and alignment with country-led priorities, while simultaneously promoting equity, innovation, and accountability in the delivery of climate finance (World Bank, 202; Climate Policy Initiative, 2021).

Popular Alternative (Climate) Finance Frameworks

Popular climate finance frameworks are aided by specific theoretical logics that influence their design, foundational assumptions, associated risks, and results (Climate Investment Funds, 2021). These frameworks innately manifest trade-offs among financial, environmental, and social priorities, highlighting the complex and often competing objectives embedded within climate finance mechanisms (Hallegatte et al., 2015). In addressing the specificities of climate finance – the increasing sophistication, distinctiveness, and idiosyncrasy, climate goals increasingly rely on innovative financial instruments and approaches that go beyond traditional aid or development finance. Alternative climate finance frameworks such blended finance, green and sustainability bonds, just transition finance, public-private partnerships (PPPs), impact investing, climate risk insurance and catastrophe bonds, and concessional finance have become central to financing the transition towards low-carbon and climate-resilient economies. These popular frameworks have gained prominence due to their relevance, scalability, institutional support, and appeal to investors (World Bank, 2021). However, these frameworks are not ideologically neutral. They are governed by a set of underlying theories and worldviews that shape their design, deployment, and performance.

Blended finance has emerged as a foundational element of climate investment strategies, facilitating the de-risking of high-risk adaptation and mitigation projects by integrating public and private capital. It combines concessional public finance with private capital to de-risk and catalyse investment in climate and environmental sectors. Rooted in neoliberal institutionalism, it assumes that market mechanisms and institutional innovation can correct capital allocation failures in developing economies. Although blended finance seeks to navigate private investment for climate goals, it may shift risks to the public sector while allowing private actors to capture the majority of returns (Convergence, n.d.). It may also divert scarce concessional finance from public goods toward bankable private ventures. For instance, IFC managed Co-Lending Portfolio Program (MCP) for infrastructure mobilizes institutional capital by blending it with concessional finance from donors (Buchner et al., 2020; Climate Policy

Initiative, 2021). Green bonds and sustainability-linked bonds have mobilized hundreds of billions of dollars globally, providing a transparent and liquid instrument for investors and governments committed to environmental, social, and governance (ESG) criteria. Guided by ecological modernization theory, these bonds are based on the idea that financial markets can be harnessed for environmental goals. It assumes that capitalist institutions can innovate towards sustainability through instruments ESG bonds (Climate Bonds Initiative, 2021). The European Union (EU) Green Bond standard and China's Green Bond catalogue are efforts to standardize and regulate this growing market. Despite increasing popularity, greenwashing risks remain substantial due to voluntary disclosure standards. There also remains sort of a tension between bond returns and long-term sustainability outcomes. Grounded in social capital theory and triple bottom line approach, impact investing is undergoing significant growth. It aligns financial returns with measurable environmental and social outcomes. It assumes that investors can align values with capital flows (GIIN, 2021). However, scaling impact investing to meet global climate finance needs remains challenging. Also, the strict impact matrices can dilute actual outcomes. Public-private partnerships (PPPs), while context-driven, have become essential for financing large-scale climate-resilient infrastructure. Following new public management and institutional economics approaches, PPP leverage private sector in delivering public services including climate infrastructure. It assumes that shared governance enhances outcomes (World Bank, 2021; Khan et al., 2020; Cochran et al., 2021; UNEP, 2021). For instance, India's Rewa Ultra Mega Solar Park project blends government support and private investment to scale renewable energy. However, in settings with limited capacity, this model may give rise to accountability gaps, cost overruns, and governance failures. Also, given competing developmental objectives, climate outcomes may be secondary to commercial returns.

Concessional finance, primarily in the form of grants and low-cost loans, remains crucial for least developed countries (LDCs) and Small Island Developing States (SIDS) to access concessional finance, provide low-interest loans or grants necessary for adaptation and mitigation finance. It is grounded in Keynesian development theory and international public finance, it emphasizes on the role of public finance – mainly from multilateral financial institutions, in stimulating environment friendly growth. (Cochran et al., 2021; Roberts & Weikmans, 2021; World Bank, 2021; UNEP, 2021; Buchner et al., 2020). The Green Climate Fund (GCF) offers concessional funding instruments such as grants, and equity guarantees for climate change adaptation and mitigation. However, as climate finance needs increase, concessional finance overrun, and strings attached to these funds may undermine country

ownership. Also, competing developmental objectives, climate outcomes may be secondary to economically struggling economies.

Frameworks such as just transition finance draws on social justice theory and political economy of climate change. It integrates equity considerations into climate investments in a way that vulnerable stakeholders benefit from green transition. (Cochran et al., 2021; Roberts & Weikmans, 2021; World Bank, 2021; UNEP, 2021). South Africa's Just Energy Transition Partnership (JETP), funded jointly by the European Union (EU), United Kingdom (UK), and United States (U.S) seeks to retire coal infrastructure while supporting affected workers and communities. Just transition interventions have gained much traction, however it is politically contested and hence remains under funded. For example, rapid decarbonization raises challenges with long-term employment and social protection needs.

Climate Risk Insurance and Catastrophe Bonds are framed under risk transfer and financial risk management theory, and disaster risks reduction (DRR) frameworks. These frameworks advocate shifting financial risks from governments and communities to private capital markets. For instance, African Risk Capacity (ARC) provides early warning focused climate insurance to African countries for drought-related shocks. While risk transfer can offer fiscal stability, it does not reduce vulnerability or help addressing underlying exposure. Also, premium affordability and low insurance penetration remain key challenges.

Overall, frameworks that integrate robust institutional support, strong investor demand, and replicability across diverse contexts. These frameworks are demonstrating the greatest scalability and popularity in both developed and developing economies.

[Table 1 in Annex-D](#) provides detailed overview of prevalent climate finance frameworks, and the factors contributing to their recognition and their comparative advantages. It also includes real-world examples from South Asia and Southeast Asia of projects that have secured funding through each framework.

Discussion on Emerging Climate Finance Frameworks

As popular alternative climate finance frameworks continue to evolve, several new frameworks have recently emerged to address prevailing gaps and limitations within the global climate finance architecture, particularly in the areas of adaptation, equity, disaster preparedness, and biodiversity (Cochran et al., 2021; Roberts & Weikmans, 2021; World Bank, 2021; UNEP, 2021; Buchner et al., 2020). Unlike established frameworks such as green bonds or blended finance, these emerging mechanisms are still in a developmental phase, characterized by

limited deployment but increasing attention from policymakers, multilateral banks, and nations vulnerable to climate impacts.

Frameworks such as debt-for-climate swaps, resilience bonds, and loss and damage finance are being shaped by pressing real-world challenges, including rising sovereign debt distress, an increase in climate-induced disasters, and a heightened demand for climate justice. Their design often incorporates innovative institutional arrangements, such as linking sovereign debt restructuring with conservation outcomes or providing compensation to nations for irreversible climate losses.

These frameworks are classified as emerging not only due to their relative novelty within the global finance landscape but also because of their experimental nature and evolving governance models. For instance, while carbon markets 2.0 and nature-positive finance draw from existing mechanisms, they prioritize next-generation standards, co-benefits, and enhanced integrity. Similarly, just transition finance represents a shift towards the integration of social equity into climate finance decisions - an area that has historically received insufficient attention.

What distinguishes these emerging frameworks is their capacity to unlock previously untapped sources of capital, direct investments towards traditionally underfunded sectors such as biodiversity and climate justice and pivot the focus of climate finance towards more inclusive and adaptive climate finance approaches (Climate Bonds Initiative, 2021; Cochran et al., 2021; Roberts & Weikmans, 2021; World Bank, 2021; UNEP, 2021; Buchner et al., 2020). While these are promising attributes, their adoption and scalability will hinge on political momentum, institutional coordination, and the capacity to demonstrate impact at scale.

The [table 2 in Annex-D](#) provides an overview of emerging climate finance frameworks, highlighting their potential and challenges for adoption. It also includes best-use case pushing these frameworks.

This paper synthesizes the key theories, models, and approaches that shape global climate change discourse and climate finance. Drawing from diverse development theories, it highlights systemic constraints and paradoxes in current climate governance. Robust theoretical grounding is essential for effective, context-sensitive climate finance strategies to achieve climate goals.

Different discourses, from environmental determinism to political ecology, shape how climate change is understood and addressed. Realism explains state-centric climate inaction, while constructivism highlights the role of shifting norms, like the adoption of net-zero targets.

Contradictory paradigms such as Marxist critiques of neoliberal finance versus ecological modernization's optimism about market innovation reflect fragmented global governance.

The mainstream market-based finance struggles with the systemic, long-term risks of climate change. Traditional finance models overlook uncertainty, vulnerability, and equity. Emerging concepts like polycentric governance and resilience theory offer alternatives, though often limited by institutional inertia and resource gaps.

Designing effective climate finance requires aligning theoretical insights with policy. Disjointed approaches risk incoherence, hence efficiency. The lack of a standard definition for climate finance further impedes policy coherence. Current frameworks often neglect enabling conditions like innovation, governance reform, and capacity building. While this review centers on theoretical foundations, it highlights limitations such as excluding deeper analysis of technological and socio-cultural factors. Future research should empirically assess how theoretical paradigms translate into effective finance mechanisms, such as loss and damage or blended finance. Interdisciplinary approaches will be key to advancing systemic, sustainable climate action. By comparing diverse theories with evolving finance mechanisms, this synthesis unveils the complexities of climate governance and the urgent need for integrated, inclusive approaches.

Conclusion

This synthesis paper offers a comprehensive review of the theoretical, conceptual, and historical underpinnings that govern the global discourse on climate change and climate finance. It contributes to the growing endeavour to reimagine both climate change discourse and global climate finance through a more integrated, justice-oriented, and forward-looking theoretical and implementation lens. Drawing upon diverse philosophical traditions, disciplinary insights, and empirical developments, this review establishes that climate change transcends a mere environmental issue, representing a deeply entrenched systemic calamity with political, economic, ethical, and epistemological implications. The theories, models, and approaches analysed in this paper not only serve as interpretative instruments but also as frameworks that inform how the issue of climate change and climate finance is delineated and whose interests are served in addressing the challenges.

Doing so transcends a mere academic exercise; it establishes an essential foundation for the formulation of responses aimed at achieving climate goals that are effective, equitable, and sustainable. This study bridges classical and contemporary theoretical paradigms with emerging interdisciplinary frameworks, examining the contested nature of climate change

governance and climate finance. It emphasizes the necessity of comprehending the multifaceted consequences of climate change as they manifest across diverse geopolitical, ecological, and economic contexts.

Synthesis insights

At the core of the paper lies a central insight: **no single theory can sufficiently capture the complexity of climate change or prescribe a universally applicable solution**. Instead, the paper demonstrates the need for an integrated and comprehensive theoretical and implementation framework—one that accommodates the diversity of human experience, scales of action, and indigenous practices. Theories such as Realism, Constructivism, Liberal Institutionalism, and Marxist Political Economy provide macro-structural interpretations of climate politics and associated finance. Meanwhile, theories like Postcolonialism, Ecofeminism, and Securitization highlight the normative, identity-driven, and often exclusionary dimensions of global climate governance. Systems Theory, Resilience Thinking, and Polycentric Governance add an ecological and institutional perspective, emphasizing feedback loops, adaptation, and multi-actor coordination.

A second major insight is that **climate finance, though a relatively recent construct in global governance, is not value-neutral**. As argued, climate finance has evolved from siloed investments to more complex frameworks emphasizing systemic change, innovation, and transformative capacity. However, dominant paradigms in finance still privilege transactional, short-term, and often technocratic responses, sidelining longer-term structural transitions, equity concerns, and contextual enablers such as the voices of the most vulnerable, indigenous practices, policy coherence, and local capacity.

Theoretical and Practical Gaps

Despite the richness of existing literature and frameworks, the synthesis identifies several critical gaps that limit the transformative potential of existing climate governance:

- 1. Epistemic Centrality of Western Theorizing:** Many dominant theories are rooted in Eurocentric traditions. This not only limits the conceptual space for indigenous, local, and non-Western knowledge systems but also shapes the priorities and practices of climate finance in ways that may not resonate with local needs or historical injustices.
- 2. The 'Theory–Practice Disjuncture':** While theories offer powerful tools for interpretation, they have limitations in articulating clear pathways to action. For example, while Postcolonial theory exposes asymmetries in climate negotiations, it does not always

provide actionable guidance for reforming financial mechanisms. Similarly, resilience theory is widely invoked in policy yet is often co-opted into neoliberal governance that shifts responsibility to communities without addressing structural vulnerabilities.

3. **Ambiguity in Climate Finance Definitions:** As noted, the lack of a standardized, transformative definition of climate finance has hindered policy coherence and implementation. Existing frameworks tend to underemphasize enabling conditions such as research, innovation, and capacity building—essential elements for achieving lasting adaptation and mitigation objectives.
4. **Blind Spots in Governance Theories:** Even well-regarded frameworks, such as polycentric governance or systems theory, often struggle with real-world implementation challenges, such as coordination failures, institutional inertia, or the politics of scale and power.
5. **Fragmentation in Institutional Architecture:** The proliferation of global, regional, and national initiatives has resulted in institutional complexity that risks duplication, inefficiency, lack of coherence, and misalignment of priorities—especially in the context of developing and Least Developed Countries (LDCs).

Recommendations and Future Research Directions

Addressing these gaps and building on this synthesis, future research must consider constructing interdisciplinary, inclusive, and actionable frameworks that inform both theory and practice. The following directions are proposed:

1. **Theorizing from the Global South:** Future work should prioritize epistemologies that emerge from climate-vulnerable regions. This includes centering indigenous knowledge systems, traditional ecological practices, and community-based finance models in global debates, thereby democratizing climate change discourse.
2. **Hybrid Frameworks for Integrated Governance:** There is a need for hybrid models that bridge normative theories, such as climate justice, ecofeminism, and postcolonialism, with operational frameworks such as climate risk disclosure and adaptation financing. Such integration would make governance more ethically grounded and practically feasible.
3. **Climate Finance Metrics and Accountability:** Further interrogation of how climate finance is monitored, evaluated, and how outcomes and learnings are reported and verified is critical. Future research should explore the politics of numbers, including how metrics (climate relevance, transformational impact, paradigm shift) are defined, who sets the criteria, and how they influence global funding flows.

- 4. Scaling Just Transitions:** Much of climate finance remains decoupled from questions of justice and transition. Future lines of inquiry should examine how financial systems can support not only emission reductions but also equitable shifts in losses and damages, energy, labor, and agriculture—particularly in the Global South.
- 5. Localizing Global Frameworks:** The gap between global-level finance and local implementation persists. Research should focus on designing mechanisms that enhance local ownership, institutional readiness, and alignment with Nationally Determined Contributions (NDCs) and National Adaptation Plans (NAPs).

Recapitulating, the insights of this synthesis paper underscore that theories are not merely tools of academic inquiry—they shape the political economy of climate action. Whether embedded in financial definitions, institutional designs, or normative framings, the theories we use have real-world consequences. As climate change accelerates and the demand for finance grows, rethinking theoretical foundations becomes not only an academic exercise but a governance necessity. Reimagining climate finance as a vehicle for enabling systemic transitions, this paper contributes to a growing body of scholarship calling for transformative, justice-oriented, and future-facing climate governance. Bridging theory, evidence, and practice is no longer optional—it is central to ensuring that climate action is not only effective but equitable, inclusive, and sustainable.

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Table 1: Overview of major global environmental conventions and agreements

Year	Convention, Treaty, Agreement	Objective(s) for development	Reasons of development
1909	Boundary Waters Treaty between USA and Canada	Addressed shared water resources between Canada and the USA	Growing concerns over cross-border water pollution and resource disputes
1946	International Convention for the Regulation of Whaling (ICRW) led by international Whaling Commission (IWC) of major whaling nations.	Regulated whale hunting due to overexploitation	Overhunting and near-extinction of several whale species
1948	International Union for Conservation of Nature (IUCN), led by governments and, UNESCO and environmental scientists.	Established one of the first global environmental organizations	Rising global concerns about biodiversity and ecosystem degradation
1971	Ramsar Convention on Wetlands, led by UNESCO, Ramsar Secretariat, wetland conservation advocates.	Focused on conserving and sustainably using wetlands	Destruction of wetlands due to urban expansion and agriculture
1972	UN Conference on the Human Environment (Stockholm Conference), led by UN (Maurice Strong – Conference Secretary General)	First global conference on environmental issues, led to UNEP creation	Industrial pollution, deforestation, and rising global awareness of environmental degradation
1973	Convention on International Trade in Endangered Species (CITES), led by UNEP, IUCN and CITIES Secretariat	Regulated international trade of endangered species due to biodiversity loss	Rapid decline in species due to unregulated wildlife trade
1979	Convention on Long-Range Transboundary Air Pollution (CLRTAP), led by UN Economic Commission for European Nations (UNECE)	Addressed transboundary air pollution affecting multiple countries	Acid rain and increasing cross-border air pollution in Europe and North America
1982	UN Convention on the Law of the Sea (UNCLOS), led by UN International Maritime Organization (IMO) and the major coastal nations	Defined ocean governance and protection of marine resources	Overfishing, pollution, and lack of maritime laws leading to resource conflicts
1985	Vienna Convention for the Protection of the Ozone Layer, led by UNEP and World Meteorological Organization (WMO)	Aimed at preventing ozone layer depletion	Scientific evidence of ozone layer depletion due to Chlorofluorocarbons (CFC) emissions

1987	Montreal Protocol on Substances that Deplete the Ozone Layer, led by UN, global scientific community	Phased out ozone-depleting substances globally	Sharp increase in skin cancer and ecological impacts due to ozone depletion
1989	Basel Convention on Hazardous Waste, led by UNEP	Regulated transboundary movement and disposal of hazardous waste	Toxic waste dumping and environmental disasters in developing countries
1992	UN Framework Convention on Climate Change (UNFCCC), led by UN, governments and global scientific community	Established global climate action framework	Increasing greenhouse gas emissions and rising global temperatures, sea levels and depleting glaciers and ice
1992	Convention on Biological Diversity (CBD), led by UN	Addressed biodiversity conservation and sustainable use	Deforestation, loss of biodiversity hotspots, and rapid species extinction
1992	UN Convention to Combat Desertification (UNCCD), led by UN's affected developing nations, desertification focused scientists	Focused on preventing desertification and promoting sustainable land use	Severe land degradation and desertification in Africa and Asia
1997	Kyoto Protocol, led by UN, Industrialized nations, and Intergovernmental Panel on Climate Change (IPCC)	Legally binding emission reduction commitments for developed nations	Growing evidence of anthropogenic climate change and international pressure to curb emissions
2001	Stockholm Convention on Persistent Organic Pollutants (POPs), led by UN and Stockholm Convention Secretariat	Banned or restricted harmful persistent organic pollutants	Chemical pollution leading to severe health and ecological risks
2009	Copenhagen Accord, led by UNFCCC, developing countries and climate finance advocates	Recognized the need for climate finance and adaptation support	Developing nations facing disproportionate climate change impacts and lack of financial support
2015	Paris Agreement (COP21), led by UNFCCC, and all signatory nations under UNFCCC	Legally binding commitment to limit global warming below 2°C	Escalating global temperatures and pressure for legally binding commitments
2015	Sustainable Development Goals (SDGs), led by UN, global policy makers, think tanks, development agencies, and academia	Integrated environmental protection into global development goals	Recognition that sustainable development must integrate environmental protection, it must be inclusive and collectively owned
2018	Katowice Climate Package (COP24), led by UN, UNFCCC, and climate negotiators	Established rules for implementing the Paris Agreement	Challenges in implementing the Paris Agreement and need for clear regulations

2021	Glasgow Climate Pact (COP21), led by UN, UNFCCC, developed and developing nations	Strengthened climate finance and mitigation commitments	Climate finance shortfalls and urgent need to strengthen global commitments
2022	Kunming-Montreal Global Biodiversity Framework, led by UN-CBD and global conservation community	Committed to protecting 30% of the planet's land and oceans by 2030	Accelerating biodiversity loss and the failure to meet previous conservation targets
2022	Loss and Damages Fund (COP27), led by UN, UNFCCC, developing nations, Small Island Developing States (SIDS), Least Developed Countries (LDCs).	Established to provide financial support to vulnerable nations facing irreversible climate impacts	Rising climate-related disasters, financial burden on developing nations, and demands for climate justice

Source: Authors research/compilation from cited sources

Table 1: Overview of major treaties and agreements on environmental governance in South Asia and Southeast Asia

Year	Treaty/Agreement	Reason for Development	Crucial Developments Triggering the Treaty, Agreement
1960	Indus Waters Treaty between India and Pakistan moderated by World Bank	Regulated water sharing between India and Pakistan	Partition of India and Pakistan led to disputes over Indus water rights
1976	Mekong River Commission (MRC) Agreement by Cambodia, Laos, Thailand, Vietnam	Established cooperation among Mekong countries for water resource management	Increased hydropower and irrigation projects in the Mekong Basin
1981	Treaty on Punjab Rivers (Ravi, Beas, and Sutlej) between India and Pakistan	Addressed water-sharing disputes over the Ravi, Beas, and Sutlej rivers in Punjab	Persistent conflicts over water allocation and increasing agricultural demand in Punjab
1995	Mekong Agreement on Cooperation for Sustainable Development among Cambodia, Laos, Thailand, Vietnam, China, Myanmar	Enhanced transboundary cooperation for sustainable development of the Mekong Basin	Environmental concerns over unsustainable use of Mekong River resources
1996	Ganges Water Sharing Treaty between Bangladesh and India	Governed water sharing between India and Bangladesh for the Ganges River	Frequent disputes over water flow from India to Bangladesh
1997	Mahakali Treaty between India and Nepal	Water-sharing and hydropower cooperation between Nepal and India	Nepal's demands for fair water allocation and hydropower development
2002	South Asia Co-operative Environment Programme (SACEP) between SAARC countries	Promoted regional cooperation on environmental protection in South Asia	Growing transboundary environmental pollution and degradation
2014	Teesta Water Sharing Agreement between Bangladesh and India	Addressed growing water disputes between Bangladesh and India over the Teesta River	Longstanding demands from Bangladesh for equitable water distribution
2015	Paris Agreement and South Asian Commitments by South Asian nations and UNFCCC	Committed South Asian countries to global climate action under the Paris Agreement	Global push for climate action and commitments from developing nations

2017	BIMSTEC Convention on Cooperation in Environment and Disaster Management among BIMSTEC nations (Bangladesh, India, Myanmar, Sri Lanka, Thailand, Nepal, Bhutan)	Strengthened regional cooperation on disaster risk reduction and environmental conservation	Frequent natural disasters and environmental degradation in BIMSTEC countries
2020	Mekong-Lancang Cooperation Agreement among China, Mekong River Commission countries	Promoted sustainable development and water management between China and Mekong nations	China's increasing influence over upstream water resources of the Mekong River

Sources: Authors research/compilation from cited sources above

Table 2: Overview of agreements and initiatives on climate finance, adaptation, and mitigation in South Asia and Southeast Asia

Year	Agreement/Initiative	Key Details
1982	South Asia Cooperative Environment Programme (SACEP), led by eight South Asian nations.	Established under Colombo Declaration, SACEP is an intergovernmental organization founded in 1982 by South Asian countries to promote and support environmental protection, management, and enhancement.
1983	International Centre for Integrated Mountain Development (ICIMOD)	Established under ICIMOD Act of Nepal, this is a Regional intergovernmental organization of eight regional member countries (RMCs) in the Hindukush Himalaya (HKH) region.
1997	The Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation (BIMSTEC), led by member nations Bangladesh, India, Myanmar, Sri Lanka, Thailand	Established under BIMSTE Declaration. Renamed as BIMSTEC Economic Cooperation. It includes climate change adaptation in its agenda for coastal and island nations
2008	SAARC Action Plan on climate change, led by SAARC member states	Established to address climate change adaptation, mitigation, and disaster risks reduction (DRR), water security, and resilience building. Limited implementation due to geopolitical challenges. Limited operation through SAARC Climate Change Centre, Bhutan
2009	ASEAN Climate Change Initiative (ACCI) led by ASEAN member states	Established to enhance regional cooperation on climate change, focusing on policy coordination and capacity building on climate change adaptation and mitigation.
2010	South Asia Climate Outlook Forum (SASCOF) led by SAARC member states.	A platform for meteorological services to develop consensus-based climate outlooks to aid adaptation planning.
2015	ASEAN Agreement on Disaster Management and Emergency Response (AADMER)	Outlined strategies for disaster risk reduction and climate change adaptation in the region. Work Programme 2016-2020 led by ASEAN member states
2015	Bangladesh Bhutan India Nepal (BBIN) Initiative	Sub-regional organization established with an aim of advancing transportation, trade and energy relationship among for member nations
2016	Asia-Pacific Climate Finance Fund (APClIFF) led by Asian Development Bank	Launched by the Asian Development Bank to support the development and implementation of financial risk management products that can help unlock capital for climate investments.
2018	ASEAN Smart Cities Network (ASCN) led by ASEAN member states	A collaborative platform aiming to develop smart and sustainable urban development to address climate change impacts.
2021	ASEAN Taxonomy for Sustainable Finance led by ASEAN member states	Established to provide a common framework to guide sustainable investment in the region, facilitating climate finance.

Source: Authors research/compilation from cited sources

Table 1: A short comparison of traditional financial theories with alternative frameworks in climate finance

Modigliani-Miller Theory - 1958: Assumes that market capital structure is irrelevant. It suggests that in perfect markets, a firm's value is independent of its capital structure (the mix of debt and equity), meaning the choice between debt and equity financing does not affect the overall value of the firm. However, in the context of climate finance, market imperfections—such as externalities, regulatory policies, and information asymmetry—affect investment in low-carbon technologies. For example: Commercial banks' reluctance to finance large-scale adaptation programs in Nepal.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Blended finance Grants Concessional loans	Uses public and donor funds to de-risk climate investments and attract private capital.	GCF using blended finance to fund Nepal's climate adaptation projects.
Capital Asset Pricing Model – Sharpe 1964: It describes the relationship between systematic or general risks of investing and expected return for assets. This model establishes a linear relationship between the return on an investment and risk. In climate finance perspective, it fails to price climate risks (physical, transition risks) and adaptation investments due to high uncertainty. For example, CAPM-based risk assessment failing to incorporate climate risks in Sri Lanka's coastal real estate.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Multi-Factor Models (Climate Risk Pricing)	Incorporates climate risks and Environmental and Social Governance (ESG) factors into asset pricing.	The European Investment Bank (EIB) integrating climate risk into project financing decisions (2020).
Efficient Market Hypothesis – Fama 1970: Assumes all risks are correctly priced and fully reflecting available information. However, climate finance market carries information gaps and pricing of adaptation benefits. For example: Carbon credit markets undervaluing adaptation benefits in sub-Saharan Africa.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Green Bonds Environmental Social and Governance (ESG) Investing	Encourages better disclosure, reporting, and market incentives for adaptation finance.	The World Bank issuing green bonds specifically for climate adaptation projects (2018).
Option Pricing Theory (Black-Scholes Model): It models capital structure decisions by viewing equity as an option on the firm's assets, and debt as a short position in a put option. Ignores real-world uncertainties and irreversibility of climate investments. For example: Commercial project financing ignoring flexibility in Bangladesh's flood-prone regions.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Real Options Theory	Provides flexibility in adaptation investment decisions based on climate risk evolution.	The UK's Thames Estuary 2100 Plan integrating real options in flood resilience planning.
Agency Theory: Misalignment between donors, governments, and local communities leads to inefficient climate finance allocation. It considers that conflicts of interest between shareholders (principals) and managers (agents) can arise, leading to suboptimal capital structure choices and firm performance. It suggests that debt can act as a governance tool to reduce these conflicts and improve firm value. For example: Large-scale misallocation of adaptation funds in India's Smart Cities program.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Just Transition Finance Stakeholder Theory	Focuses on equitable, inclusive, and transparent climate finance governance.	The EU's Just Transition Fund supporting coal-dependent communities (2020).

Pecking Order Theory: It assumes companies prioritize financing from internal funds in an order of retained earnings and debt followed by new equity. Issuing equity is seen as a negative signal, suggesting overvaluation. Debt is preferred as it provides a less ambiguous signal to the market. This theory implies no optimal capital structure, and financing decisions depend on the company's financing. However, adaptation projects often lack immediate revenue streams. For example: Commercial firms in Nepal rejecting adaptation investments due to uncertain revenue streams.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Concessional Finance De-Risking Instruments	Encourages adaptation finance by offering risk-sharing and low-cost financing options.	The IFC's concessional finance support for climate adaptation in Bangladesh.
Market Failure Theories: Market failures, such as information asymmetry, agency costs, and the presence of externalities, significantly influence capital structure decisions leading to suboptimal investment choices. Public goods nature of climate adaptation leads to underinvestment by the private sector. For example: Lack of adaptation investment from private sectors in Nepal's water security		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Public-Private Partnerships Catalytic Finance	Creates structured incentives to involve private investors in adaptation.	The Asian Development Bank (ADB) supporting PPPs for climate resilience in the Pacific Islands.
Mean-Variance Portfolio Theory (MPT), also known as Modern Portfolio Theory: Focuses on financial risk diversification but ignores long-term climate resilience factors. It is a framework for constructing investment portfolios that maximize returns for a given level of risk, focusing on diversification and the trade-off between expected returns and variance (or standard deviation). For example: Mutual funds avoiding long-term investments in climate-vulnerable areas in India.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Climate-Resilient Investment Portfolios	Optimizes asset allocation by integrating long-term climate adaptation metrics.	The Norwegian Sovereign Wealth Fund adjusting portfolios to include climate resilience metrics
Liquidity Preference Theory: Investors prefer liquid assets, making long-term adaptation projects unattractive. It assumes that companies' capital structure decisions are influenced by the market's preference for liquid assets. For example: Institutional investors avoiding long-term adaptation investments in Nepal.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Adaptation Bonds Long-Term Climate Finance Instruments	Introduces fixed-income financial products designed for adaptation financing.	The African Development Bank (AfDB) issuing adaptation bonds for infrastructure resilience (2019).
Arbitrage Pricing Theory (APT): It assumes that the value of a firm's capital structure depends on factors that affect asset returns and how sensitive those returns are to various risk factors. It does not account for climate-related financial risks and uncertainty. For example: Standard risk pricing failing to account for flood risks in Indonesia.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Climate Risk-Adjusted Asset Pricing	Develops adaptation risk premiums in asset pricing models.	The TCFD (Task Force on Climate-Related Financial Disclosures) incorporating climate risks into financial reporting (2017).
Intertemporal Capital Asset Pricing Model (ICAPM): It helps explain capital structure by considering how investment and financing decisions are made across multiple time periods. Unlike the static CAPM, ICAPM acknowledges that investors are concerned with changes in future investment opportunities, not just current wealth. It does not consider changing climate-related investment		

opportunities over time. For example: Financial institutions ignoring future climate projections in South Asia.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Scenario-Based Climate Finance Models	Models' financial decisions under various future climate scenarios.	The Network for Greening the Financial System (NGFS) integrating scenario-based climate stress tests (2020).
Prospect Theory & Loss Aversion: It assumes that individuals feel the pain of a loss more strongly than the pleasure of an equivalent gain. This loss aversion instinct limits to take on debt (perceived as riskier and potentially leading to losses) and instead prefer equity financing, even if debt might be cheaper or more tax efficient. This behavior stems from a desire to avoid potential downside and preserve existing wealth leading to suboptimal financial decisions. In climate finance perspective, investors and policymakers discount long-term adaptation risks in favor of short-term gains. For example: Policymakers in Nepal prioritizing immediate infrastructure over long-term adaptation.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Behavioral Finance for Climate Risk Perception	Improves risk communication and incentives for long-term climate investments.	The EU is introducing climate risk disclosure regulations to change investor perceptions.
Stakeholder Theory: It suggests that firms should consider the preferences of all stakeholders, not just shareholders, when making capital structure decisions, recognizing that these decisions impact various groups like employees, customers, and the community. In climate finance perspective, decisions often exclude vulnerable communities. For example: International adaptation finance often bypassing local governance in Nepal.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Community-Based Climate Finance	Ensures local ownership and inclusiveness in adaptation finance planning.	The Global Environment Facility (GEF) supports local-led adaptation in LDCs.
Sustainable Finance Theory: Sustainable finance theory explores how capital structure decisions can be aligned with environmental, social, and governance (ESG) goals, suggesting that companies with better ESG performance may have access to cheaper capital and a more favorable capital structure. Traditional financial models do not integrate ESG and sustainability factors. For example: Limited sustainability-linked financial products in South Asia.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Sustainability-Linked Finance	Aligns financial incentives with environmental and social outcomes.	The Singapore Exchange mandating sustainability reporting for listed firms (2021).
Real Options Theory: Traditional models assume static investment decisions without flexibility. analyzes the value of managerial flexibility to adapt and revise future decisions to capitalize on favorable future investment opportunities or to limit downside losses from adverse market developments, which is vital to long-term corporate success in an uncertain and changing marketplace. For example: Traditional rigid investment frameworks in climate-vulnerable regions of Nepal.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Adaptive Climate Investment Strategies	Encourages flexible, phased investment in adaptation projects.	The US Federal Reserve is exploring real options-based approaches for climate resilience financing.
Value at Risk (VaR) Model: It is used to assess the level of risk associated with various types of assets in the financial domain. It does not account for long-term, nonlinear climate risks. For example: Banking stress tests failing to assess long-term climate risks in South Asia.		
Alternative Framework	How it bridges the gap	Examples of alternative framework

Climate Stress Testing & Scenario Analysis	Incorporates extreme weather risks into financial decision-making.	The Bank of England implementing climate stress tests for financial institutions (2021).
Information Asymmetry Theory: It implies that one party possessing more information than the other, significantly impacts capital structure decisions, leading firms to potentially choose debt over equity and potentially increasing financing costs. Investors lack climate-related financial data, leading to inefficient markets. For example: Greenwashing in corporate sustainability reporting in emerging economies.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Climate Disclosure Transparency Initiatives	Improves market efficiency through enhanced climate risk reporting.	The SEC (US) proposes mandatory climate risk disclosures for publicly traded firms (2022).
Trade-off Theory: Assumes firms balance debt and equity financing based on tax benefits and financial distress costs. In the context of climate finance, climate investments require more concessional and flexible financing. For example: Limited climate investment due to high debt servicing costs in developing economies.		
Alternative Framework	How it bridges the gap	Examples of alternative framework
Concessional Finance (Grants & Low-Cost Loans).	Reduces the cost of climate investments, making them more attractive by offering lower interest rates and reducing financial distress risks.	Green Climate Fund's concessional finance for climate-resilient infrastructure in Nepal (2022)

Table 1: Popular climate finance frameworks

Framework (initiators)	Reasons for popularity	Comparative advantage	Examples
Blended Finance (OECD and World Economic Forum, DFIs, IFC, ADB, AfDB, GCF, Rockefeller Foundation (Convergence Blended Finance)).	De-risks adaptation investments, attracts private capital, enables large-scale climate finance combining concessional and commercial funding. Growing due to increased international support from multilateral organizations.	More scalable than public grants, lower risk than pure private finance, encourages private sector participation in adaptation projects. Ensures long-term financial sustainability.	GCPF India Blended finance for renewable energy (2021)
Green Bonds, and Sustainability Linked Bonds (Climate Bonds Initiative, World Bank initiated first green bond in 2008, IFC, Asian Infrastructure Investment Bank).	A transparent, fixed-income climate finance instrument, appealing to institutional investors and ESG driven markets. Growing due to rising demand for sustainable investments.	Low-cost capital, increased investor confidence due to sustainability commitments, encourages accountability, and expands funding sources beyond traditional finance.	Indonesia's Green Sukuk Bond for climate projects (2021)
Just Transition Finance (European Bank for Reconstruction and Development, Just Transition global platforms like COP26/COP27).	Encourage climate policies do not disproportionately impact vulnerable communities, aligning climate action with social justice. Supported by major climate negotiations (COP agreements).	Combines climate action with labour and economic transition support, secures funding for social protection in the green transition. Encourages policy alignment and social equity.	ADB's ETM in the Philippines Coal phase-out with labour support (2021).
Public-Private Partnerships (World Bank, Governments, UNDP for climate -aligned PPP toolkits)	Mobilizes private capital and technical expertise to develop resilient infrastructure while ensuring long-term operational sustainability. Supported by government and multilateral agencies to enhance adaptation infrastructure.	Considered to be more efficient than standalone government funding, allows shared risk between public and private sector, creates sustainable financing models, increases financial viability.	Bangladesh's Climate-Resilient Infrastructure PPP (2021)

Impact Investing (Global Impact Investing Network, Rockefeller Foundation, Social enterprise funds, and development finance institutions)	Aligns capital with measurable climate and social impact, attracting socially responsible investors seeking both returns and sustainability outcomes. Increasing due to a shift in investor preferences towards ESG and impact metrics.	More targeted than blended finance, integrates financial viability with environmental goals, encourages long-term sustainable investments. Expands adaptation finance sources beyond conventional funds.	LeapFrog Investments in India Climate-smart agribusiness (2021)
Climate Risk Insurance and Catastrophe (Cat Bonds) (Munich Re, Swiss Re [reinsurers], African Risk Capacity, Southeast Asia Disaster Risk Insurance Facility, Caribbean Catastrophe Risk Insurance Facility, World Bank Treasury)	Reduces financial risks from climate disasters, provides pre-emptive infrastructure financing for disaster-prone regions, and stabilizes economies post-disaster. Gaining traction due to increased climate-induced disasters.	Transfers risks to private markets, provides rapid liquidity after disasters, avoids the need for post-disaster emergency relief dependency. Strengthens financial resilience against extreme climate events.	SEADRIF's Sovereign flood risk insurance in Laos (2020)
Concessional Finance (Grants and Low profit Loans) Multilateral Development Banks, GCF, LDCF, Adaptation Fund, Bilateral donors)	Provides affordable capital for climate action, particularly in LDCs and SIDS. Critical for financing projects with high social returns but low financial returns.	Reduces financial barriers for vulnerable economies, supports early-stage project development, enables long-term financial sustainability without creating excessive debt burdens.	Green Climate Fund's concessional finance for climate-resilient infrastructure in Nepal (2022)

Source: Authors research/compilation from cited sources

Table 2: Emerging climate finance frameworks

Emerging Frameworks	Scalability Potential				Overall adoption and scalability	Potential challenges for adoption/ scalability	Best use cases
	Market demand	Policy support	Financial feasibility	Early success			
Debt-for-Climate Swaps	☑☑ High	☑☑ Strong	☑ Moderate	☑☑☑ Strong case studies	☑☑ High	Requires strong governance and bilateral negotiations	In 2021, Belize restructured \$553 million of external debt through debt-for-nature Swap to protect 30% of its marine area
Carbon Markets 2.0	☑☑☑ Very High	☑☑ Strong	☑☑ High	☑☑☑ Strong case studies	☑☑☑ Very High	Needs better regulation and transparency to prevent greenwashing	Singapore launched a carbon credit hub (Green Plan 2030), funding verified tropical forest conservation.
Resilience Bonds	☑☑ High	☑☑ Strong	☑☑☑ Very High	☑ Few case studies	☑☑ High	long project cycles make it unattractive for some investors	In 2022, Indonesia issued resilience bonds to finance coastal defence against sea level rise.
Nature-Positive Finance	☑☑ High	☑☑ Strong	☑ Moderate	☑ Few case studies	☑☑☑ Very High	Still lacks clear financial mechanisms	In 2023, India launched blue carbon credits to finance mangrove restoration.
Loss and Damage Finance	☑☑ High	☑ Moderate	☑ Moderate	☑ Moderate	☑ Early stages	Politically sensitive, slow to implement. Undefined mechanisms.	Germany, UAE, EU pledged over \$700 million to kick-start the fund.
Finance in Common (FiCS)	☑☑ High	☑ Moderate	☑☑☑ Very High	Ø Evolving	☑ Vague Pledges	Currency risks, power asymmetries, economic disorders	FiCS 2025, MDB and CFIs reimagining sustainable finance – as a new normal.

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